

Natural Disasters, Industrial Policy, and Innovation: Evidence from the Great Chicago Fire *

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Abstract

This paper examines whether, in response to natural disasters, industrial policy can shape innovation toward risk-mitigating technologies. We study the 1871 Chicago Fire and the consequent policy banning wooden construction. Using a triple differences framework, we find that construction innovation and manufacturing in Chicago increased sharply, with positive spillovers into related sectors. To distinguish the effects of the Fire and the policy, we compare wood and non-wood construction, showing that gains were concentrated in non-wood construction. Additionally, we study the 1872 Boston Fire, where no regulations were implemented, and find modest effects on construction patenting and manufacturing.

Keywords: Great Chicago Fire, Industrial Policy, Directed Innovation.

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1 Introduction

Natural disasters cause significant damage worldwide, and with ongoing climate change, these adverse effects are expected to intensify over the coming decades (Cruz and Rossi-Hansberg, 2024). Adaptation hinges on technical change (Barreca, Clay, Deschenes, Greenstone and Shapiro, 2016). Directed innovation in US agriculture, for example, has offset roughly 20% of the potential losses from damaging climate trends since 1960 (Moscona and Sastry, 2023). Despite these prospective gains, economic theory indicates that market competition may lead to inefficiently low levels of innovation in areas that will generate new products or technologies in the future (Acemoglu, 2012, 2023).¹

In this paper, we investigate whether public policy can address this inefficiency by guiding the innovation response to natural disasters and supporting technologies that mitigate their adverse effects. We examine how innovation and manufacturing responded to the 1871 Great Chicago Fire and to the subsequent legislation that prohibited wooden constructions within the city borders. Over the second half of the nineteenth century, Chicago witnessed momentous growth owing to its central location within the US railway network. This unregulated sprawl left it exposed to fire hazards. Estimates indicate that 100,000 people—out of a population of 300,000—were left homeless by the 1871 Fire, which caused approximately 200 million dollars in damages (5 billion 2010\$), or 670\$ per inhabitant (Smith, 2020). Following the Fire, the Chicago municipal authority passed an ordinance in 1874 prohibiting the construction of wooden buildings within the city’s borders. In the following decades, construction practices in Chicago introduced key innovations, including fireproof materials, iron frames, and the first skyscrapers. The “Great Rebuilding” attracted many architects and engineers, and Chicago emerged as the global hub of innovative architecture (Condit, 1952).

A simple approach to identify the causal effects of the fire would be to compare Chicago to control cities before and after 1871. This strategy, however, requires that Chicago and the control cities would have followed similar trajectories in the absence of the fire. This assumption is unlikely to be verified, given that Chicago was the fastest-growing city in the U.S. and the main hub of the expanding railway network (Miller, 1996). To address this challenge, we employ a triple-differences design that compares changes in innovation and manufacturing activity in construction-related sectors relative to other sectors, before and after the 1871 Fire, relative to a set of control cities. Relative to a more traditional difference-in-differences estimator, this triple differences strategy controls for time-varying city-level shocks common across sectors, sector-specific time-varying unobserved heterogeneity common across cities, and differences in initial sectoral specialization across cities. We assess the empirical plausibility of the parallel differences assumption through event-study specifications, which reveal no statistically significant differential pre-Fire trends between treated and control groups (Olden and Møen, 2022). As a complementary approach, we show that the results are robust to a synthetic control method which leverages information on Chicago and untreated cities to construct a “synthetic”

¹Acemoglu (2012), for instance, uses electric and internal-combustion cars as an example. At time t , internal-combustion cars have higher quality and are thus preferred by consumers, but in $t' > t$, consumer tastes will change, and they will buy electric cars. Since private returns to innovation in internal combustion technology will be higher than in electric cars, market competition underprovides innovation in the “clean” technology.

Chicago that serves as a counterfactual to estimate the effects of the Fire (Abadie and Vives-i Bastida, 2022).²

We find that the 1871 Fire had a positive and large impact on construction-related innovation in Chicago. Between 1871 and 1900, the number of construction patents issued to inventors living in Chicago increased by 20% relative to control cities and sectors. This wedge widened over the twenty years that followed the Fire and decreased thereafter. Overall, approximately 48 additional patents in construction were granted in Chicago relative to other sectors and cities over the 30 years that followed the Fire. To estimate the knowledge spillovers of the increased construction innovation, we measure the technological similarity between construction and other technology classes.³ Patenting in Chicago increased relative to control cities in classes technologically closer to construction, while it decreased in more distant technology classes. Our estimates thus indicate a reallocation of innovation activity towards construction technologies. Moreover, the different trajectories across sectors suggest that the response of innovation to the Fire is unlikely to be due to general economic growth spurred by the rebuilding of Chicago. If this were the case, we would observe similar increases in innovation across all technology classes, while we find a much stronger effect in innovation in construction-related sectors.

Turning to broader indicators of economic activity, we use the value of production, number of establishments, capital, and material and labor costs from manufacturing censuses, as digitized by Hornbeck and Rotemberg (2024). The 1871 Fire impressed a major upward shift on construction manufacturing in Cook County, where Chicago is located. Relative to other cities and sectors, the number of construction establishments and their output in Chicago increased by 38% within a decade of the Fire. Production value, fixed capital, material, and labor costs show even higher growth over the same period.

Third, we examine the impact of the 1871 Fire on historical landmark buildings as an additional indicator of innovation in construction technologies. We assemble a geo-referenced dataset covering the universe of sites listed in the National Register of Historic Places (Stutts, 2024). We compare the effect of the Fire on architectural landmarks to all other buildings. As discussed in Section 2, the “Great Rebuilding” ushered in an unprecedented agglomeration of architects and engineers in Chicago. We find that the number of architecturally relevant landmarks in Chicago approximately tripled relative to control cities by 1880 and further increased until 1900. The presence of these architectural sites, deemed worthy of preservation, arguably proxies for the architectural innovations developed in those years. By contrast, we find a considerably smaller increase in the number of non-architectural landmarks in Chicago after the Fire.

Did the “industrial policy” intervention implemented by the Chicago municipality affect innova-

²Within the historical urban economics literature, the synthetic control method has been recently applied by Becker, Heblich and Sturm (2021) to study how changes in public employment caused by the designation of Bonn as the capital of the German Federal Republic after World War 2 impacted private-sector economic activity.

³We adopt two measures of cross-sector technological similarity. First, we rank sectors by the share of patents we identify as construction-related. Second, we adopt a text-based similarity measure that leverages document embeddings (Mikolov, Sutskever, Chen, Corrado and Dean, 2013). The two approaches yield very similar results.

tion and economic activity? We provide two pieces of evidence to answer these questions.

First, we examine how the impact of the 1871 Chicago Fire differed between wood-related and non-wood-related innovation and manufacturing, within the construction sector. Since wood-related construction was forbidden within the city limits, the construction policy plausibly redirected the demand shock, generated by the Fire, into non-wood construction. Our results corroborate this interpretation. Non-wood construction innovation accounts for most of the increase in overall construction innovation: by 1890, Chicago-based inventors obtained about 20% more non-wood construction patents relative to inventors based in control cities, whereas we find no statistically significant increase in wood-related patenting. The effects of the Chicago Fire on non-wood manufacturing indicators further support this pattern. The number of non-wood construction establishments and their production value, respectively, doubled and increased by 80% in the ten years after the 1871 Fire. Material and labor costs also display substantial increases. The results hold when controlling for total employment, further suggesting that the increase in patenting and manufacturing is driven by the construction sector. These patterns are consistent with our hypothesis that Chicago's construction policy channeled reconstruction efforts into non-wood construction, thus propelling innovation in that area. Non-wood buildings pose significantly fewer fire hazard concerns and permit denser agglomeration. In hindsight, non-wood construction would become the predominant technology, vastly outpacing wood construction and dominating contemporary urban landscapes.

Second, we examine the impact of the 1872 Boston Fire on construction innovation and manufacturing. In Boston, the fire destroyed part of the central business district, causing 75 million dollars in damages (1.8 billion 2010\$), or 300\$ per inhabitant, which corresponds to 44% of the damages caused by the Chicago Fire, in per capita terms (Hornbeck and Keniston, 2017). Public opinion called for more restrictive legislation, similar to the one enacted in Chicago, but lobbying by real estate developers successfully opposed any initiative in this direction. These different institutional constraints resulted in vastly different reconstruction experiences. In Boston, post-Fire buildings largely resembled those that preceded them. Although the Boston and Chicago fires differed in proportions, they are still comparable in many dimensions. They were both severe, targeted the central business districts, and hit economically dynamic and growing cities. We thus view the 1872 Boston Fire as an instructive counterfactual, where no post-Fire construction policy is implemented. We thus replicate the analysis focusing on the 1872 Boston Fire. The triple-difference estimates indicate that the Boston Fire had a statistically significant positive effect on construction innovation, but its magnitude was less than 10% of the increase we estimate in Chicago. Similarly, we find considerably more modest increases in economic activity in construction in Boston than in Chicago, especially in non-wood construction, where Boston exhibits no increase after the 1872 Fire relative to control cities.

The differences between wood and non-wood innovation and manufacturing, as well as the results of the 1872 Boston Fire, provide consistent, suggestive evidence that construction policies implemented in Chicago influenced manufacturing and innovation dynamics in the aftermath of the fire.

The Great Chicago Fire provides a unique natural experiment to examine how public policy can

direct endogenous innovation responses to natural disasters and help mitigate their adverse effects. Our results reveal substantial scope for policy interventions to sustain resilience-enhancing innovation amid impending climate-related challenges.

Contributions to the Literature This paper contributes to three streams of literature. First, we contribute to the literature on the direction of innovation. Pioneering studies by Habakkuk (1962) and Schmookler (1966) and subsequent theoretical contributions by Acemoglu (2002, 2010) show that market size and relative factor prices shape the direction of innovation and have received vast empirical support (e.g., Popp, 2002; Hanlon, 2015; Andersson, Karadja and Prawitz, 2022; San, 2023). Recent papers document how natural disasters influence the direction of technological change using cross-country variation (Miao and Popp, 2014), as well as focusing on specific events, such as droughts (Moscona, 2025), climate change (Moscona and Sastry, 2023), and epidemics (Berkes, Coluccia, Dossi and Squicciarini, 2025). Our contribution here is twofold. First, we disentangle the effect of policy interventions from other, more general, post-Fire consequences and provide novel evidence that policy intervention can successfully direct the endogenous innovation response to natural disasters and mitigate their adverse consequences. Second, we show that technological change responds to the damages caused by urban fires, a growing concern amid contemporary global warming and overpopulation.

Second, we add to the growing literature on the economics of industrial policy (Juhász, Lane and Rodrik, 2023; Bartelme, Costinot, Donaldson and Rodriguez-Clare, 2025). Recent papers study the efficacy of deliberate industrial policy interventions in sustaining industrialization in textiles (Juhász, 2018), shipbuilding (Kalouptsi, 2018; Hanlon, 2020), heavy chemicals (Lane, 2025), among others, as well as their effects on regional development (Incoronato and Lattanzio, 2024; Garin and Rothbaum, 2025; Mitrunen, 2025). Additionally, the study of innovation policy has gathered considerable attention (Bloom, Van Reenen and Williams, 2019), particularly in terms of public R&D (e.g., Azoulay, Graff Zivin, Li and Sampat, 2019; Gross and Sampat, 2023; Moretti, Steinwender, Van Reenen and Warren, 2025). We inform this literature by building on Acemoglu (2012), who argues that market competition generally under-provides diversity in innovation. Our findings provide the first evidence that innovation policy can decrease excess conformity to the incumbent technology and increase the amount of diversity in technological change.

Third, we contribute to the literature on the economic effects of natural disasters. Existing papers adopt either a cross-country perspective (Dell, Jones and Olken, 2012; Cattaneo and Peri, 2016; Kocornik-Mina, McDermott, Michaels and Rauch, 2020) or focus on specific disasters, such as the 1871 Boston Fire (Hornbeck, 2012), the 1906 San Francisco Earthquake (Ager, Eriksson, Hansen and Lønstrup, 2020), the Dust Bowl (Hornbeck, 2012), the 1927 Mississippi Flood (Hornbeck and Naidu, 2014), and the 2009 Ketsana typhoon in Vietnam (Gröger and Zylberberg, 2016). In between these two approaches, Boustan, Kahn, Rhode and Yanguas (2020) construct a long dataset of natural disasters in the United States to employ within-country identifying variation. Strobl (2011) and Mahajan and Yang (2020) adopt a similar approach focusing on hurricanes, whereas Borgschulte, Molitor and Zou (2024) study wildfires. We present mixed evidence on the effects of natural disasters *per se*, while

shedding light on the key role of policy intervention following such shocks.⁴

Outline of the Paper Section 2 presents a description of the Chicago and Boston fires. In Section 3, we describe the data used in the analysis. Section 4 discusses the causal research design, and in Section 5, we present the main results on the effects of the 1871 Chicago Fire. Section 6 explores the underlying mechanisms and discusses the effect of the 1872 Boston Fire. Section 7 concludes.

2 Historical Background

This section describes the events of the Great Chicago Fire of 1871, the construction policies that followed it, and their consequences on construction practices and technological advancements. Then, we present key information on the Great Boston Fire of 1872.

2.1 The Great Chicago Fire of 1871

The Fire started on October 8, 1871, and ravaged central Chicago for two days. Estimates indicate that hundreds of individuals died, and close to 100,000 people were left homeless following the conflagration (Smith, 2020). More than 16,000 buildings, worth approximately 200 million dollars (or 5 billion dollars at current prices), chiefly in the central business district, were destroyed.⁵ The Fire devastated infrastructure, including water works, railroads, and private buildings, such as hotels and theaters.

Two factors, in particular, determined the devastating impact of the Fire. First, Chicago had been massively growing in the decade preceding the Fire through the construction of wooden slums which spread from the city's outskirts deep into the center (Rosen, 1986). Miller (1996) documents that the lack of regulatory oversight was instrumental in creating fire hazards in the central portion of the city. Second, the Fire Department proved ineffective in containing the Fire (Smith, 2020). Various alarms raised by the citizens failed to be communicated to the fire station. Firemen, who were badly equipped, faced logistical problems as the chaos in the streets obstructed their intervention.

2.2 Construction Policy After the Fire

The pre-Fire growth of the city of Chicago was not accompanied by urban policy oversight. In 1871, only one piece of legislation prevented the construction of wooden buildings in a small area of the city center without affecting existing housing stock (Rosen, 1986). Soon after the 1871 Fire, a debate arose around the necessity to pass more stringent legislation on new constructions. Joseph Medill, elected in 1871 shortly after the Fire, enacted minor provisions affecting relatively small portions of the new immigrant neighborhoods. Only in 1874, after a second—minor—Fire threatened the center, the National Board of Fire Underwriters (now American Insurance Association), an organization established in 1866 by insurance companies to reduce fire losses and promote fire safety, successfully pushed for more comprehensive legislation to prevent further disasters (Critchell, 1909).

⁴On the one hand, the 1872 Boston Fire had a modest positive effect on innovation and economic activity. Conversely, the 1871 Chicago Fire had substantial positive effects on both variables.

⁵According to the 1870 census, the population of the inner Chicago area was approximately 300,000. The contemporary 200 million 1871\$ figure equaled approximately 2.6% of the US GDP, which, today, would imply 700–800 billion 2010\$ damages.

The first provisions enacted by the Medill administration shortly after the 1871 Fire allowed owners in each block to vote and decide to outlaw buildings made of inflammable materials. The ordinance issued after the 1874 fire, by contrast, prescribed that only bricks and non-combustible materials would be used in new buildings. Frame buildings, the predominant construction technology in Chicago, were prohibited except for very small units far from the city center. Warehouses and lumber yards were removed from all built-up areas within the limits of Chicago. Moreover, the fire department was reorganized and furnished with new equipment (Smith, 2020). The National Board of Fire Underwriters served as guarantor of the implementation of the legislation enacted by the municipal administration. While direct evidence on the enforcement of the ordinance is not available, the Board received a large number of requests to amend the law for specific wooden buildings, all of which were rejected by the Council (Mayer, Wade, Holt and Pyle, 1969).⁶

2.3 Construction Practices and Innovations after the Fire

Despite initial fear that the new restrictive ordinance would hamper the city's redevelopment, rebuilding efforts were swift (Rosen, 1986). Compared to the pre-Fire buildings, the newly erected constructions had thicker walls, deeper foundations made of mortar and brick, often encapsulated elevators, and featured better fire escapes. They were also larger in area and taller, thus resulting in a larger internal space. The 1871 Fire ultimately ushered in substantial innovations in building technology and design, attracting many architects and engineers who would make Chicago a global hub for innovation in construction technology—a legacy that persists to this day (Wermiel, 1996).

The first breakthrough innovations were in fireproof construction. While fireproof techniques had been developed long before the Fire, their adoption and further advancement drastically accelerated after 1871. George H. Johnson, a designer who moved to Chicago in 1871, patented a hollow-tile system for fireproof floors (Sawislak, 1995). The application of terra-cotta tile to cover exposed iron parts became widespread. Being heat resistant, the tile remained intact in the heat of direct flames, thus providing a key step toward the design of fireproof buildings (Peters, 1991).

Under the ordinance's provisions, new constructions had to rely on an iron frame instead of a wooden inner structure. As land prices in the central business district area soared, developers sought to erect increasingly tall structures (Condit, 1952). Eventually, the first skyscrapers in history emerged in Chicago. In 1895, a writer in *Engineering News* cited in Randall (1949) reports that "The construction of enormously high office buildings [...] originated in Chicago, in its practical application, at least, and that city has at the present time buildings of steel skeleton type than have all other American cities together." The development of skyscrapers required substantial innovation in iron frames to reduce their weight and improve their stability. Additionally, the development of tall buildings promoted the fast adoption of the elevator.⁷

The "Great Rebuilding" attracted many architects, engineers, and designers. Historians refer to this newly formed community as the "Chicago School Architecture" (Fitch, 1948). The guiding prin-

⁶The requests appear in the Proceedings of the Common Council of the City of Chicago in 1874, 1875, and 1876.

⁷The Home Insurance Building, commonly considered the first skyscraper in the world due to its steel-framed structure, was inaugurated in Chicago in 1885. It had four elevators serving its ten floors.

ciples of this heterogeneous movement—which embraced structural innovations such as steel-frame construction and improved wind resistance while also stressing the importance of spacious interiors and abundant light (as in Holabird and Roche’s 1889 Tacoma building design)—reverberated in European Modernism and contributed to making Chicago the global hub of innovative architecture (Condit, 1952).

2.4 The Great Boston Fire of 1872

The Fire in Boston broke out on November 6, 1872. It centered in the wholesale business district and destroyed 776 buildings, causing approximately \$75 million in damages, or 11% of the total assessed value of Boston real estate stock (Hornbeck and Keniston, 2017). Post-fire growth was fueled by strong private demand and resulted in growing land prices and capital inflows.

As in Chicago, public opinion called for stricter building regulations. Facing lobbying by powerful interest groups, however, the municipal authority could only enact weak building legislation, which was ultimately repealed in 1873 (Rosen, 1986).⁸ Thus, unlike in Chicago, the reconstruction in Boston was not overseen by the municipal authority and was largely privately managed.

Partly because of the different policy constraints faced by developers in Boston and Chicago, post-fire buildings in Boston largely resembled those that preceded them. Fireproof techniques were applied to traditional materials, and the construction techniques remained largely unchanged. Rosen (1986) argues that this approach was rooted in the structural differences between the two cities. Boston had already undergone a rationalization of land use before the Fire, and consequently, its urban infrastructure was less amenable to dramatic improvement. As a result, Boston’s post-fire reconstruction produced safer and better fireproofed buildings, yet largely unchanged in construction methods and architectural style.

3 Data

This section presents the data underlying the analysis and our procedures for compiling the final datasets. First, we present a new dataset of patents issued between 1853 and 1900. Then, we discuss how we use the manufacturing and population censuses and describe a newly compiled dataset of geo-coded historical US landmarks.

3.1 Patent Data

We measure innovation activity using patents, in line with a long tradition in economics (Griliches, 1990). We collect the text of the universe of patents issued in the United States between 1853 and 1900 from Google Patents, following the approach of Moser and San (2020).⁹ Using a state-of-the-art

⁸In Chicago, landowners and industrialists were advocating for tighter regulations, whereas the less affluent feared that these regulations would force them out of the city center. On the other hand, in Boston, landowners advocated against tighter regulations because the city had already undergone extensive redevelopment before the fire (Rosen, 1986). These different positions partly explain the different provisions enacted in the two cities.

⁹The United States Patent and Trademark Office (USPTO) was established in 1836. While data on patents issued before 1853 exist, patenting was rare and unusable for our empirical analysis.

large language model (GPT 4o-mini), we extract, directly from the patent’s text, data on the inventors’ name, address, location, the filing and issuance date of the patent, and potential firm assignee. We augment this information with data on the patent CPC technology classes provided by Google Patents. Furthermore, using a commercial software (Google Maps API), we geo-code the addresses extracted from the patent documents to precise latitude and longitude coordinates.¹⁰

We identify construction-related patents using a simple dictionary-based approach. We label a patent as “construction” if it mentions at least five times one or more construction-related words (in a sample of 30 words). The list of construction-related words is provided in Table D.1. Thus, patents that the USPTO did not assign to the “Fixed Construction” technology class may still be classified as construction. We adopt a similar method to identify “wood-related” and “non-wood-related” construction patents. In particular, we label a patent as “wood-related” (resp. “non-wood-related”) if: (i) it is a “construction” patent, and (ii) it mentions one or more wood-related (resp. non-wood-related) words within a pre-designed dictionary. On average, Table I shows that 17% of patents are flagged as related to construction. Of those, 12% are related to wood-related construction, while 23% are identified as non-wood-related construction patents.

To quantify the technological spillovers of the Chicago Fire on innovation outside of construction, we measure the technological similarity between construction-related and other patents by technology class. We propose two alternative procedures. First, we rank CPC classes by the share of patents in each class, which are also construction patents. Second, we employ the doc2vec document embedding model, a natural language processing technique (Mikolov et al., 2013). The model is trained on a 20% random sample of the universe of patents, allowing us to represent each patent as a real-valued vector. For each non-construction patent, we compute the average cosine similarity with all construction patents and take the average within each technology class. Thus, the ranking returns a technology class-level order of similarity between construction and non-construction patents. The two methods return the same ranking: the most similar technology classes are classes G (“Physics”), Y (“General Tagging of New Technological Developments”), and B (“Performing Operations; Transports”). By contrast, patents in classes D (“Textiles, Paper”) and C (“Chemistry; Metallurgy”) are the least similar to construction patents. Technology classes F (“Mechanical Engineering; Lighting; Heating; Weapons; Blasting”) and A (“Human Necessities”) are in the middle tercile of the similarity distribution. We exclude patent class H (“Electricity”) because very few patents in electricity were issued before 1871.

Using the coordinates assigned to the location data extracted from patent documents, we assign patents to the locations listed in the Census Place Project (CPP), a directory of geo-coded locations listed in the US census (Berkes, Karger and Nencka, 2023). Specifically, we assign a patent to the closest CPP location provided that at least one of the inventors resides within 20 kilometers (12.4 miles).¹¹ The results are not sensitive to alternative thresholds between five and thirty kilometers.

¹⁰We compare our newly constructed data to Sarada, Andrews and Ziebarth (2019) and Petralia, Balland and Rigby (2016), which cover a partially overlapping time period but do not contain information on the text of patent applications. Our data coverage exceeds 90% of both datasets for the years when they overlap (see Section A for details).

¹¹We assign patents with multiple inventors to each inventor’s CPP location. Results remain unchanged if we assign equal

3.2 Manufacture Census

We use county-by-industry data from the 1860, 1870, and 1880 Censuses of Manufactures digitized by Hornbeck and Rotemberg (2024). The data contain information on production value, number of establishments, value of fixed capital, and labor and material costs. The analysis using data from the Censuses of Manufactures is run at the county level. We thus assume that the Chicago Fire affects the entire Cook County (IL), where Chicago is located. Similarly, the Great Boston Fire impacted Suffolk County (MA), where Boston is located.¹² Since the data is at the decade level, we use 1880 as the only post-treatment period, whereas 1860 and 1870 constitute the pre-treatment window.

Table I lists selected statistics for the Manufacturing Census data. Counties have an average of ten establishments, although this figure conceals substantial heterogeneity. Cook County, where Chicago is located, and Suffolk County, where Boston is located, host more than 1,000 firms, placing them in the top percentile of the overall distribution. Other economic performance indicators, such as the total production value, display similar degrees of heterogeneity.

We consider the “construction,” “construction materials,” and “furniture” sectors as “construction manufacturing.” Our baseline results remain unchanged if we only include the “construction” sector in the treatment group. Within these sectors, we refer to the finer taxonomy produced by Hornbeck and Rotemberg (2024) to identify wood- and non-wood-related industries. Wood-related industries are those labeled as “carpentering,” while non-wood-related industries are “brick, stone, and tile,” “marble and stone work,” and “lime and cement.”

3.3 Population Census

We use individual-level data from decennial population censuses and location data from the CPP (Ruggles, Fitch, Goeken, Hacker, Nelson, Roberts, Schouweiler and Sobek, 2021; Berkes et al., 2023). The CPP assigns precise latitude and longitude coordinates to the large majority of individuals in the population census, which lacks standardized and comprehensive location data beyond the county of residence. We use these two sources to produce two distinct datasets.

First, we tabulate data from the 1870 population census at the CPP-location level. For each location, we compute a set of demographic characteristics listed in Table I (Panels B–D). Among those, we choose the set of variables we use to construct the synthetic control units.

Second, we construct an individual-level dataset from the 1870 and 1880 censuses. We consider the universe of the working-age population—i.e., above 15 years old—with a valid occupational response. We identify as a construction worker any individual listing an occupation where at least 50% of the employed are listed in the “Construction” industry in the federal census. Among those, “Carpenters” are identified as wood-related construction workers, whereas we assign “Brickmasons, stonemasons, and tile setters,” “Cement and concrete finishers,” and “Plasterers” to non-wood-related construction. On average, 1.9% of the workforce is employed in construction, 1.1% is employed in

shares to each inventor’s location.

¹²This assumption is reasonable, for Chicago’s and Boston’s metropolitan areas account for more than 90% of the population of Cook and Suffolk Counties in 1870.

wood-related construction jobs, and 0.3% is assigned to non-wood-related construction occupations.

3.4 Historical Landmarks

To assess the cultural and architectural legacy of the Chicago Fire, we assemble data on significant buildings erected in the metropolitan areas in our sample between 1850 and 1900. We start from places listed in the “National Register of Historic Places” (Stutts, 2024). The National Register is the United States’ official list of historically significant places. It was established under the 1966 National Historic Preservation Act and is currently maintained by the National Park Service. The Register records buildings, districts, sites, and, more generally, places deemed worthy of preservation.

The Register lists 99,199 sites.¹³ It indicates the state, county, and city where each entry is located, along with the type of record (e.g., “district” or “building”), the area of significance (e.g., “architecture” or “industry”), and other information less relevant for the analysis. We first geo-code each entry and assign it to the closest metropolitan area within 20 kilometers (12.4 miles). We successfully geo-coded 98% of the Register. We find that 24,673 sites (24.8% of the sample) are assigned to a metropolitan area in our sample. Second, we augment the dataset with information on the construction year of the landmarks. To do so, we individually search for each entry on Wikipedia. We correctly impute a construction year to approximately 80% of the entries.

The final sample of historical landmarks erected in one of the metropolitan areas in the sample, along with information on their construction year, comprises 19,110 entries. In the analysis, we concentrate on the subset of 5,907 landmarks erected between 1850 and 1900. Additionally, we focus on 3,792 “buildings” entries, thus discarding residual observations listed as “district,” “site,” and “object.” Moreover, we distinguish units based on their area of significance: “architecture” and all others. Examples of non-architectural landmarks include Native American heritage sites, homes of prominent political and civic figures, natural sites, locations of significant historical events such as battles, and industrial heritage sites.

3.5 Construction of the Samples

The first sample we construct is a city-level panel dataset where patenting activity is observed every five years between 1855 and 1900. The data covers the largest *metropolitan areas* in the United States in 1870. To construct a metropolitan area, we first extract all CPP locations with at least 20,000 individuals in the 1870 census.¹⁴ There are 84 such places, each corresponding to a major city. Then, we map all other minor towns to the closest city with a population above 20,000, provided their distance does not exceed 20 kilometers (12.4 miles).¹⁵ The results remain qualitatively unchanged for thresholds between 5 and 30 kilometers. The resulting dataset thus comprises 84 “metropolitan areas” which include a single major town above 20,000 inhabitants and all other minor towns within 20 kilometers

¹³We accessed the Register in October 2024, when the last update was dated August 2024.

¹⁴Our results remain robust when we apply alternative population thresholds between 5,000 and 50,000 to construct the donor metropolitan area pool. We focus on relatively large cities primarily because the synthetic control approach applies a zero weight to small towns when constructing the synthetic control, as they are too different from the treated cities.

¹⁵Figure C.2 provides a graphical description of the agglutination procedure that forms the “Chicago” metropolitan area within Cook County (IL).

from its center.¹⁶ For each metropolitan area, we tabulate demographic characteristics from the 1870 census—the last census before the Fires—and compute the number of patents issued between 1855 and 1900. Figure I displays the geographic distribution of all metropolitan areas thus constructed.

The second sample is constructed at the county level and comprises all counties where our metropolitan areas are located. There are 76 such counties. The difference between the number of counties (76) and metropolitan areas (84) is due to the fact that some counties encompass more than one metropolitan area.¹⁷ These data are available at a decennial frequency between 1860 and 1880. For each county, we observe data from the Census of Manufactures and information extracted from the 1870 population census and mapped onto 1870 county borders. Data from the 1860 and 1880 Census of Manufactures are cross-walked to county borders in 1870, following the methodology described by Eckert, Gvirtz, Liang and Peters (2020). We observe the number of establishments, production value, fixed capital, material costs, and labor costs in construction, wood-related manufacturing, and non-wood-related manufacturing.

Third, we complement the individual-level data extracted from the 1870 and 1880 population censuses with the intergenerational links produced by Abramitzky, Boustan, Eriksson, Pérez and Rashid (2020). We link the 1870 entries to their records from the 1880 census so that, for each individual, we observe a set of fixed individual characteristics and time-varying outcomes—specifically their industry and occupation—at two points in time, 1870 and 1880. Because we seek to measure how the Chicago and Boston fires affected the probability of taking jobs in construction, we restrict the sample to include individuals between 15 and 70 years old with a valid occupational response in either census, who were not working in construction-related occupations in 1870.

4 Research Design

Our empirical analysis seeks to evaluate the impact of the 1871 Fire on innovation and manufacturing activities. The key challenge is that other correlated shocks in Chicago may affect the variables of interest. A natural approach would thus be to compare Chicago with other cities in a difference-in-differences setting. This strategy, however, relies on the hypothesis that without the Fire, Chicago and the other cities would have followed similar trajectories. This parallel trends assumption is unlikely to be verified. Before the Fire, Chicago was the fastest-growing large city in the United States and a central hub of the expanding railway network (Miller, 1996). It thus seems plausible that Chicago was not on the same trend as other cities before 1871.

To address this concern, we employ a triple-differences design that exploits variation across cities, sectors, and time. The estimator compares changes in innovation and manufacturing activity in construction-related sectors relative to other sectors, before and after the Fire in Chicago, relative to a set of control cities. By differencing along these three dimensions, the design absorbs (i) time-varying city-level shocks that affect all sectors within a city; (ii) sector-specific shocks that are common across

¹⁶The full list of cities above 20,000 inhabitants, as well as all minor cities within their metropolitan areas, is provided in Table D.2.

¹⁷In robustness exercises, we show that the baseline results also hold in a larger sample comprising all US counties.

cities, such as technological change or demand shifts; and (iii) time-invariant differences in sectoral specialization across cities.

Formally, we estimate variants of the following regression:

$$f(\mathbb{E}[y_{ikt}]) = \alpha_{it} + \alpha_{kt} + \alpha_{ik} + \sum_{\tau=-a}^b \beta_{\tau} \cdot I(i = \text{Chicago}) \cdot I(k = \text{Construction}) \cdot I(t - 1871 = \tau), \quad (1)$$

where i , k , and t respectively denote a city (or county), sector, and time period. In the baseline specification, we control for city-by-time (α_{it}), sector-by-time (α_{kt}), and city-by-sector (α_{ik}) fixed effects. The treatment is a triple interaction that compares Chicago and the construction sector to other cities and sectors over time since the 1871 Fire. The functional specification $f(\cdot)$ depends on the outcome variable y_{ikt} . For patents and landmarks, we employ a Poisson pseudo-likelihood estimator that reflects the fact that they are count variables. For outcomes constructed from the census of manufactures, we employ a linear specification and express y_{ikt} in log terms.¹⁸ We also report the results of a pre-post specification analogous to (1), where we replace the event-time dummies with a post-treatment indicator $I(t \geq 1871)$.

The identifying assumption underlying this strategy is that, in the absence of the Fire, the relative evolution of construction and non-construction sectors in Chicago would have mirrored that in the control cities. Intuitively, while Chicago may have been on a distinct aggregate trajectory, driven, for example, by its rapid population growth or its role as a transportation hub, such factors are unlikely to have differentially affected construction-related activities relative to other sectors in a way that diverged systematically from other cities. As long as unobserved shocks influencing the sectoral composition of economic activity evolved similarly across cities, the triple-differences estimator nets out Chicago’s idiosyncratic growth dynamics and isolates the differential impact of the Fire on construction-related innovation and manufacturing.

We assess the plausibility of this parallel differences assumption through event-study specifications, which show no evidence of differential pre-Fire systematic differences across treated and control sector–city pairs, as discussed in more detail in the next section.

As a complementary approach, we employ a synthetic control estimator (Abadie and Gardeazabal, 2003; Abadie, Diamond and Hainmueller, 2010). The core idea of synthetic controls is to employ information on treated and control units to construct a “synthetic” control that “resembles” the treated unit and can serve as a counterfactual.¹⁹ The estimated causal effect is, thus, the difference between the treated and synthetic control outcome values after the treatment period. Relative to the triple-differences estimator, the synthetic control approach identifies the treatment effect from the extent to which Chicago’s post-Fire outcomes deviate from the path predicted by its own pre-treatment dynamics, rather than from cross-sector contrasts. As such, the synthetic control estimates capture the aggregate effect of the Fire and the associated reconstruction policies on the outcome of interest,

¹⁸We view the Poisson specification as most appropriate for patent variables in light of Chen and Roth (2024), who show that more traditional zero-inflated log-linear models yield average treatment effects that are neither scale invariant nor point identified.

¹⁹Specifically, we match on pre-treatment outcome values and population, literacy, income per capita, share of white inhabitants, share of natives, and employment shares by occupation and industry, all measured in 1870.

under the assumption that the weighted combination of control cities provides a valid approximation to Chicago’s untreated counterfactual. Overall, the synthetic control and triple difference methods yield qualitatively consistent results.

5 Main Results

This section presents the main results of the paper. We organize it into four parts. First, we explore the effect of the Chicago Fire on innovation in construction. Then, we look at how the effects of construction innovation spilled over into innovative activities in other industries. Third, we explore the broader consequences of the Fire on the manufacturing sector. Finally, using data on historically significant landmarks, we document the historical and cultural legacy of the Fire.

5.1 The Great Chicago Fire and Innovation in Construction Technology

The central objective of this paper is to understand how innovation responded to the 1871 Fire shock, shedding light on the role of policy interventions. To quantify this effect, we first focus on construction-related technological change, as measured by patents. We contrast construction patenting with patenting in other sectors.

We begin with a graphical comparison of raw means. In Figure II, the red line corresponds to Chicago, while the gray line denotes the average across control cities. The first panel reports the number of patents in construction-related technologies at five-year intervals, whereas the second panel reports patenting activity in non-construction technology classes. In both panels, outcomes are normalized to one in the last pre-Fire period (1865–1870).

Prior to the Fire, Chicago and the control cities exhibited broadly similar trends in both construction and non-construction patenting. In the post-Fire period, however, the two groups diverge sharply in construction-related innovation. Fifteen years after the Fire, construction patenting in Chicago exceeds its pre-Fire level by more than a factor of thirteen, while control cities experience a five-fold increase. By contrast, the divergence is substantially smaller in non-construction technologies. Over the same horizon, non-construction patenting in Chicago rises to approximately four times its pre-Fire level, compared to a twofold increase in control cities.

Figure IIc displays the triple-differences event-study coefficients associated with (1), which can be interpreted as the double difference between Chicago and the control cities, and between construction and the control sectors, for each given period. We find no evidence of statistically significant pre-treatment differences between treated and control groups. The increase in construction patents increases for up to fifteen years following the fire and decreases thereafter. Over the years 1880 to 1885, Chicago produced approximately 35% more patents in construction than in other sectors compared to control cities.

Table II reports the results obtained from a pre-post specification. In column (1), we report the coefficient from a difference-in-differences regression comparing the number of patents in construction in Chicago and in control cities. The positive coefficient we obtain, however, may reflect broader growth in Chicago rather than the specific effect of the Fire on construction innovation. To address

this concern, column (3) reports estimates from the triple-differences specification, which additionally exploits within-city comparisons across technology classes to difference out city-specific time-varying shocks. The triple-differences estimate implies that, on average over the post-Fire period, the number of construction-related patents increased by approximately 20 percent in Chicago compared to non-construction technologies relative to the control cities. In levels, this corresponds to roughly 48 additional construction patents attributable to the Fire before 1900. This figure is approximately one quarter of the difference-in-differences estimate, suggesting that broader economic growth in Chicago had a positive effect on construction patenting, in addition to the effect of the Fire.

Patents vary extensively in economic significance. The standard practice to account for this heterogeneity is to look at citations. This approach is infeasible in historical settings because citations of prior art became compulsory only after World War II (Andrews, 2021). To address this limitation, we adopt the novelty indicator developed by Kelly, Papanikolaou, Seru and Taddy (2021), which measures novelty as the excess text similarity of each patent with future patents relative to previous patents. In columns (2) and (4), we repeat the analysis focusing on the number of patents in the top 20% of the novelty measure distribution. The positive effect of the Fire on construction innovation is not driven by low-novelty innovation, as we estimate an even larger effect of the Fire on high-novelty patenting activity, even though, in this case, the magnitude (a 63% increase) is driven by the low mean of the dependent variable.

A plausible concern is that the estimated treatment effect for Chicago is not uniquely driven by the Fire, but instead reflects a broader pattern whereby similar post-treatment effects would emerge for other cities if assigned the same shock. To explore this possibility, in Figure III, we display the distribution of the average treatment effects we obtain when assigning the “treated” status separately to each city. The solid black line denotes the estimate for Chicago, which lies at the 95th percentile of the resulting distribution. This indicates that the estimated effect in Chicago is large relative to placebo cities, suggesting that the increase in construction patenting is unlikely to be driven by spurious city-specific dynamics rather than by the Fire itself.

In Figure VIIa, we report the synthetic control estimates. Specifically, we assign the treatment status to each of the 84 metropolitan areas in the sample and compute the difference between construction-related patenting in that city and its associated synthetic control. This difference is an estimate of the treatment effect for Chicago and for “placebo” units. We highlight the estimates for Chicago in red, while all other metropolitan areas are shown in gray.²⁰ The treatment response in Chicago far exceeds all other cities. If we pretended that the 1871 Fire had happened in any other city, we would never estimate a response of construction-related innovation as big as it is in Chicago.

Figure C.5a further confirms that the results remain qualitatively unchanged when employing the synthetic difference-in-differences estimator developed by Arkhangelsky, Athey, Hirshberg, Imbens and Wager (2021), which embeds a synthetic control matching framework in a double differences

²⁰Following standard practice in synthetic control estimation, we exclude placebo cities whose mean squared prediction error (MSPE) is much larger (ten times) than Chicago’s MSPE. Practically, this only excludes New York.

estimator.²¹

Additionally, Figure C.3a indicates that the statistical significance of the estimates remains unchanged when employing alternative standard error estimators, including the Conley (1999) correction to account for the possible presence of spatial autocorrelation. Figure C.4a further shows that the baseline result is not driven by any specific city included in the control group, as the coefficient estimate remains stable when excluding one city at a time from the estimation sample.

Altogether, these results suggest that the 1871 Fire triggered an increase in construction-related innovation in response to both the need for reconstruction and policies promoting the use of fireproof and innovative designs and materials.

5.2 Knowledge Spillovers

The previous analysis provides strong evidence that the 1871 Fire propelled a self-sustaining wave of innovation in construction in Chicago that spanned at least three decades after the Fire. In this section, we examine how the boom in construction innovation spilled over to other fields depending on their technological similarity to construction. This exercise is informative along two dimensions. First, it allows us to gauge the relevance of knowledge spillovers in sustaining subsequent innovation. Second, we can use fields other than construction to construct a “within-Chicago” counterfactual. By comparing the pattern of innovation activity across sectors within Chicago, one can net out the aggregate impact of the Fire and disentangle its effect on construction.

Our text-based measure of “construction” patents does not exploit technology class information. Thus, construction patents are scattered across CPC technology classes. Our main measure of technology-class-level similarity to construction is the share of patents in any CPC technology class (except construction) flagged as “construction” by the text-based algorithm. Then, we rank classes according to this share and divide them into terciles such that the bottom tercile comprises classes with the least share of construction patents, and the top one collects those with the largest share.

Following the previous analysis, we estimate the treatment effects of the Fire, using as the outcome variable the number of non-construction patents by tercile of similarity with construction patents. Figure IV reports the results of this exercise. The dot reports the triple differences treatment effect of the Fire of construction innovation. The red bars report the regression coefficients obtained using the number of patents issued in technology classes that are increasingly distant from construction as the outcome variables.

We find that the effect of the 1871 Fire on patenting activity is positive in sectors that are technologically closest to construction. In those sectors, the effect is comparable in magnitude to the increase in construction innovation itself. As technological distance from construction increases, this relationship reverses. Patenting in the middle tercile of technological distance exhibits no statistically significant response to the Fire, while patenting in the most technologically distant sectors declines (by 25%, on average). This gradient is consistent with a reallocation mechanism whereby the surge

²¹See Section B.2 for a more detailed discussion of the synthetic difference-in-differences estimator and the methodology we employ to construct the standard errors displayed in the event-study results.

in construction-related innovation generated knowledge spillovers benefiting technologically proximate sectors, while drawing innovative activity away from more distant technology classes.

The different trajectories of patenting across terciles of similarity to construction innovation suggest that the response of construction innovation to the Fire is unlikely to be due to general economic growth spurred by Chicago's rebuilding. If this were the case, one would expect similar increases in innovation in other technology classes. Figure IV, instead, clearly indicates that patenting in technologies distant from construction did not significantly increase—and, in fact, decreased—after the Fire.

The synthetic control estimates reported in Figure VIIb reveal a similar pattern. Prior to 1871, patenting in Chicago closely tracks that of its synthetic counterpart across all terciles of technological similarity to construction. Following the Fire, the trajectories diverge sharply, with the largest post-1871 increases concentrated in sectors most closely related to construction.

5.3 The Impact of the Fire on Construction Manufacturing

Patents provide a direct way to quantify the economic response to the Great Chicago Fire at a high time frequency and over a long period. In this section, we examine the broader economic implications, focusing on the manufacturing sector.²² Specifically, we compute the total number of establishments, the value of production, the value of installed fixed capital, and labor and material costs of establishments based in one of the 76 counties where the original 84 metropolitan areas are located. All variables are expressed in log, and we employ a linear specification.

Figure V reports event-study estimates based on decennial county-level data from 1860, 1870, and 1880, with 1880 constituting the post-Fire observation. Across all outcome variables—except for the number of establishments—we find no evidence of differential pre-Fire trends between construction manufacturing in Cook County, where Chicago is located, and other counties or sectors. Following the Fire, construction manufacturing activity in Chicago increases sharply relative to non-construction sectors and to construction activity in other counties.

Panel A of Table III reports pre-post estimates of the effect of the Fire.²³ We find large increases across all outcomes. Relative to other sectors and cities, the value of production in construction in Cook County more than doubles, while labor costs and installed capital increase by 95 percent and 111 percent, respectively. The number of establishments in construction increased by 38%, and the value of installed capital rose by 112%. Although these magnitudes partly reflect the relatively small size of the construction sector in Cook County prior to 1870, they nonetheless indicate that the Fire triggered a substantial expansion of construction-related manufacturing activity.

How does the effect of the Fire on construction compare to that on other manufacturing sectors? Figure VI addresses this question by reporting estimated treatment effects on the value of production obtained by sequentially assigning “treated” status to each manufacturing sector, using the remaining

²²The two main disadvantages relative to the previous analysis are: (i) manufacturing data are from decennial censuses taken between 1860 and 1880, and (ii) they are county-level tabulations instead of the city-level data constructed from patents.

²³Panel A of Table D.3 reports the analogous estimates obtained on the full sample of US counties, irrespective of whether they contain one of the metropolitan areas in the sample.

sectors as controls.²⁴ The results indicate that construction manufacturing in Chicago experienced the fastest growth in the post-Fire period. The second-largest increase occurs in Iron and Steel, a sector plausibly complementary to construction activity. Taken together, this pattern suggests that the Fire disproportionately accelerated activity in construction and closely related industries, consistent with the view that the shock reshaped Chicago's industrial structure during the years of the "Great Rebuilding."

Figure C.7 replicates the analysis employing the synthetic control framework at the county level. The synthetic control attains a good pre-Fire fit for each county, as the difference in patenting between treated and (synthetic) control counties is close to zero across all outcomes. Except for the number of establishments, Cook County stands out as the unit that deviates the most from the synthetic control in 1880, after the Fire. The synthetic control results are thus consistent with the estimates obtained from the triple differences specification.

The estimated effect of the Chicago Fire on construction activity remains statistically significant when employing alternative standard error estimators (see Figure C.8). In addition, the estimates remain stable when excluding one county at a time from the estimation sample (see Figure C.9), which indicates that the effect is not driven by comparing Cook County with selected areas in the United States.

5.4 A Quantitative Analysis of the Cultural Legacy of the Fire

We conclude our analysis of the effects of the 1871 Great Chicago Fire by studying its cultural legacy. We measure the cultural impact of the Fire using comprehensive data on historical buildings listed in the National Register of Historic Places. We compute each metropolitan area's total number of sites by construction year. We view landmarks as a measure of the economically relevant innovations in construction practices and technologies. The National Register aims to list all places deemed worthy of preservation due to their cultural importance. Hence, a count of sites indicates the cultural legacy of the Fire and its impact on architecture. As we discussed in Section 2, there is vast qualitative evidence that the Fire was a crucial agglomeration factor for architects and designers who ultimately formed the first Chicago School of Architecture (Wermiel, 1996).

We begin by presenting raw means for landmark construction in Chicago and in control cities. Figure C.10a plots the average number of architectural landmark buildings in Chicago (red) and in control cities (gray). Architectural landmark designation increases sharply in Chicago, with the number of landmarks in 1900 exceeding its pre-Fire level by more than a factor of four. By contrast, no comparable upward trend is observed in control cities.

It is important to note that, to some extent, these divergent trajectories may reflect the mechanical replacement of structures destroyed by the Fire. We thus examine non-architectural landmarks, that is, buildings designated as of national significance for reasons other than their architectural merit. In Figure C.10b, we find no difference between Chicago and control cities prior to the 1871 Fire, and no differential change after 1871: the number of non-architectural landmarks remains relatively stable

²⁴Figure C.6 reports the results for all other manufacturing outcomes.

in both groups. Taken together, these patterns suggest that the post-Fire increase is specific to architectural landmarks rather than a reflection of a general change in landmark designation rules or the destructive effects of the Fire, since the sample only covers buildings.

With this caveat in mind, we report the Poisson regression estimates in Table IV. Since landmark construction is relatively rare, we group years by decade. Columns (1–3) display the estimates from a double-differences regression where we compare Chicago to control cities. In column (1), the dependent variable is the overall number of landmark sites, whereas in columns (2) and (3), we distinguish between architectural and non-architectural sites. Column (4) presents the result of a triple-differences specification, which compares architectural and non-architectural landmarks in Chicago and in control cities before and after the Fire.

We find a sizable increase in the number of listed buildings in Chicago after the Fire relative to the counterfactual. The total number of landmark buildings exhibits a 235 percent increase relative to the pre-Fire average (column 1). However, the picture is very different when contrasting architecture and non-architecture sites. The increase in historical landmarks is largely driven by the growth of architectural landmarks relative to the pre-Fire average. On average, we estimate that the number of architectural landmarks displays a three-fold increase in Chicago compared to control cities after the 1871 Fire (column 2). By contrast, the number of non-architecture-related sites increases by approximately 90 percent (column 3). The triple differences regression displayed in column (4) compares the two categories and, as in the previous exercises, allows us to include city-by-time fixed effects to net out the increase in landmark construction that is driven by Chicago’s growth. Estimates from this more demanding specification indicate that architectural landmark sites increased in Chicago relative to control cities and non-architectural sites by more than three times after the Fire. This sharp difference echoes the historical evidence indicating that the Fire was a decisive factor in making Chicago the center of US architecture throughout the second half of the century. Provided that culturally relevant heritage sites partly reflect innovations in construction technology and techniques, these patterns provide evidence that the pace of construction innovation in Chicago greatly accelerated after the 1871 Fire.

Our results provide evidence that the Great Chicago Fire fostered construction manufacturing and technological advancements. These patterns are consistent with historical narratives documenting the swiftness of the “Great Rebuilding” of the city over the following decade. At the same time, however, they also indicate that these effects were not short-lived, as they shaped economic activity in the area well into the twentieth century. Chicago did not die out of economic distress following the Fire. Instead, it grew to be the second-largest city in the United States. Our results thus indicate that economic responses to adverse natural disasters may reverse their adverse effects into opportunities for growth. In the next section, we explore the role of the “industrial policy” intervention in shaping the magnitude and characteristics of such a large response.

6 Mechanism

This section examines potential mechanisms that may explain the results presented thus far. We begin by studying the effects of the Chicago Fire on wood- and non-wood-related construction innovation and manufacturing to try to isolate the impact of construction policies from the effects of the Fire as such. Second, we explore the employment dynamics in Chicago before and after the Fire using longitudinal linked census data. Finally, we use the 1872 Great Boston Fire as a laboratory to study the effects of urban fires in the absence of significant construction policy interventions.

6.1 Non-Wood and Wood Construction Innovation after the Fires

We begin by studying how the innovation response to the 1871 Chicago Fire was influenced by the policy that forbade the construction of wooden buildings within the city perimeter. The historical literature suggests that the policy played a key role in shifting economic activity toward non-wood construction (see Section 2.2). Table V reports the results. Columns (1), (3), (5), and (7) present double-difference estimates comparing the number of non-wood and wood-related patents in Chicago to the control cities, while columns (2), (4), (6), and (8) report triple-difference estimates that compare non-wood and wood-related innovation in Chicago to other sectors and cities before and after the 1871 Fire.²⁵

In columns (1–4) and (5–8), we examine non-wood and wood construction patents separately. Column (1) documents a large post-Fire increase in non-wood construction patents in Chicago. Quantitatively, the estimate implies an approximately threefold increase relative to the control cities, corresponding to about 40 additional patents. Column (3) focuses on high-novelty patents and reveals an even larger response. The coefficient implies that, of the total additional non-wood construction patents granted in Chicago after the Fire, about 12 meet the high-novelty criterion proposed by Kelly et al. (2021). Column (5) reports the effect on wood construction innovation. We estimate a positive post-Fire increase in the number of wood construction patents, though the magnitude is smaller than for non-wood construction.

A potential concern is that wood and non-wood construction technologies may have been on systematically different innovation trajectories, in which case the estimates could reflect a secular shift toward non-wood construction rather than the effect of the Fire. Similarly, the double-difference estimates may reflect broader economic growth in Chicago after the Fire, rather than the effect of the Fire on non-wood and wood construction patenting activity. To assess these possibilities and provide more solid causal evidence, in columns (2) and (4), we estimate a triple differences regression where we compare non-wood patenting to other technologies and cities to net out city-specific time-varying unobserved factors as well as time-varying technology-specific shocks that may explain surging patenting. Columns (6) and (8) replicate this exercise for wood construction patenting.

The diverging trajectories of wood- and non-wood construction innovation are particularly clear

²⁵In the triple differences regressions, when looking at non-wood (resp. wood) construction, we exclude wood (resp. non-wood) construction patents from the sample. Additionally, we exclude the residual “construction” patents that are neither wood- nor non-wood related.

in the triple-difference estimates. Column (2) shows that non-wood construction patenting increased by 20% percent relative to other sectors in Chicago compared to control cities, while the effect on high-novelty patents (column 4) is even larger. By contrast, we find no increase in wood construction innovation relative to other sectors in Chicago, compared to control cities, after the Fire, both in terms of the overall patenting activity (column 6) and the number of high-novelty patents (column 8).

The large difference between wood and non-wood construction innovation is consistent with our hypothesis that the policy intervention enacted by the Board of Fire in 1874 to forbid wooden constructions within the city perimeter shifted the demand for new buildings onto non-wood constructions. This sharp demand shock ushered in momentous innovation in non-wood construction, ultimately contributing to the overall increase in construction innovation.

In Figure VIIc–Figure VIId, we report the synthetic control results for non-wood and wood construction innovation, which provide additional evidence in support of the difference-in-differences estimates. Compared to the synthetic control unit, Chicago (in red) exhibits no differential trends in patenting in either non-wood or wood construction before 1871. After the Fire, however, Chicago sharply diverges from the synthetic control, and the increase in non-wood construction patenting is more than twice as large as the increase in wood construction innovation. Importantly, Chicago stands out as an outlier among all other placebo cities in the sample (in gray).

Lastly, in Figure C.5b and Figure C.5c, we display the estimates obtained using the synthetic difference-in-differences method, which yield a qualitatively consistent picture. Non-wood and wood construction innovation increased in Chicago after 1871, but the increase in the former is considerably larger—about twice as large—than the increase in wood-related construction technologies.

6.2 Non-Wood Construction Manufacturing after the Fires

Following the logic of the previous section, we now turn to the effects of the 1871 Fire on manufacturing in non-wood construction. Since innovation and broader economic activity should co-move, we expect innovation in non-wood construction manufacturing to increase. In this part of the analysis, we do not look at wood construction industries because the census of manufactures only records carpentering—the wood-related construction sector—in 1860. As in Section 5.3, all outcomes are expressed in log-terms and estimates are obtained using a linear specification.

Table VI reports the results from a difference-in-differences specification that compares non-wood construction manufacturing in Cook County with control counties before (in 1860 and 1870) and after (in 1880) the Fire.²⁶ The estimates indicate an increase in non-wood construction manufacturing activity across all outcomes except capital. Quantitatively, the number of establishments in non-wood construction roughly doubled after the Fire in Chicago relative to control cities, implying that 5 additional firms operating in non-wood construction were established by 1880 in Chicago. We find similar positive effects of the Fire on other indicators of non-wood construction manufacturing activity. The value of production of firms in non-wood construction almost doubled, and we find similar increases

²⁶The baseline results are obtained on the sample of counties with at least one metropolitan area within their boundaries. Panel A of Table D.4 replicates the analysis on the sample of all counties in the United States.

for labor and material costs.

Figure C.11 reports the synthetic control estimates, where we compare non-wood manufacturing in Cook County to the synthetic Cook County control unit. Across all outcomes we consider, Chicago displays the largest—or second largest, in capital and material cost—deviation from the synthetic control among all counties in the sample, thus providing further evidence suggesting that the increase in non-wood construction we estimate in Cook County, where Chicago is located, was likely caused by the Fire and, plausibly, by the construction regulations that followed it.

As in the rest of the analysis, we document that the effect of the Fire on non-wood construction activity is robust to employing alternative estimators for the standard errors (Figure C.12) and to excluding one county at a time from the estimation sample (Figure C.13), thereby confirming that our results are not sensitive to spatial autocorrelation and are not driven by specific counties in the control group.

These patterns provide further evidence of increased economic activity in construction and the growing relevance of the non-wood construction sector, and are consistent with our hypothesis that the regulation that banned wooden buildings in post-Fire Chicago redirected economic activity towards this sector.

6.3 Employment Dynamics after the 1871 Fire: Evidence from Linked Individual-Level Data

In this section, we leverage the granularity of the census data to measure employment transitions into construction, wood-, and non-wood-related construction in Chicago. We construct employment from decennial population censuses and link individuals between the 1870 and the 1880 waves using state-of-the-art algorithms developed by Abramitzky et al. (2020). Because we look at individuals in the labor force, we exclude those without a valid occupational response from the sample. Our guiding question is thus to understand whether individuals living in Chicago at the time of the Fire were more likely to take up construction jobs and, more specifically, whether these jobs are more likely to be in non-wood *vis-à-vis* wood construction.

We estimate variations on the following difference-in-differences specification:

$$y_{it} = \alpha_{c(i,t)} + \beta_t + X_i' \Gamma + \delta \times [I(c(i,t) = \text{Chicago}) \times I(t = 1880)] + \varepsilon_{it}, \quad (2)$$

where i denotes an individual, $t \in \{1870, 1880\}$ denotes a census wave, and $\alpha_{c(i,t)}$ and β_t denote the city where inventor i lives in year t and census-wave fixed effects. The term X_i collects a set of individual-level controls—race, birth year, literacy, migration status, and broad occupational category—measured in 1870.²⁷ The explanatory variable of interest is the treatment defined as the interaction between a dummy equal to one for individuals residing in Chicago either in 1870 or in 1880 ($I(c(i,t) = \text{Chicago})$) and a post-Fire dummy variable ($I(t = 1880)$). Standard errors are clustered at the city level. We employ three dependent variables: construction employment, wood-related construction employment, and non-wood-related construction employment.

Under a standard parallel trends assumption, the difference-in-differences coefficient $\hat{\delta}$ captures

²⁷Results are robust to controlling for individual fixed effects (α_i).

the causal effect of living in Chicago after the 1871 Fire on the probability of working in construction. Since we observe individuals only twice and census data are available at a decennial frequency, we cannot rule out that the estimates partly reflect shocks other than the Fire occurring in Chicago that influence the outcome variables. We thus view this exercise as providing descriptive, but not necessarily causal, evidence on the labor market effects of the Fire.

Columns (1–3) of Table VII report the results. In column (1), the dependent variable is an indicator equal to 100 if the individual works in construction in 1880 and zero otherwise; in column (2), we restrict the attention to non-wood-related construction occupations; in column (3), we only consider wood-related construction jobs. Our estimates indicate that treated individuals are 1.8% more likely to work in construction in 1880. This shift is large in magnitude as it corresponds to approximately 120% of the mean. This large effect is plausibly consistent with urban reconstruction in the aftermath of the destructive effects of the 1871 Fire. In column (2), we show that treated individuals were approximately 0.23% more likely to work in non-wood-related construction, implying that the share of individuals employed in non-wood construction in Chicago doubled relative to the mean. By comparison, in column (3), we find that while wood-related construction employment increased after the Fire, this effect corresponds to a 50% increase relative to the mean. These findings echo evidence from the Census of Manufactures, indicating that construction manufacturing activity in Chicago increased after the Fire and that such an increase was primarily (though not exclusively) driven by non-wood-related industries.

The heterogeneous response of wood and non-wood construction to the 1871 Fire documented using individual micro-data, is thus consistent with more aggregate—yet more solidly causal—city- and county-level evidence and indicates that the booming construction industry in Chicago after the Fire was predominantly dominated by non-wood construction firms, as envisioned by the 1874 municipal ordinance.

6.4 The Impact of the 1872 Great Boston Fire on Construction Innovation and Manufacturing

A possible explanation for the pattern of construction manufacturing in Chicago after the Fire is that it reflects a more general economic upswing impressed by the demand shock originating from the rebuilding effort, rather than the effect of the industrial policies implemented. We already saw that this explanation is unlikely: first, the heterogeneous responses of innovation to the 1871 Fire depended on the relative similarity to construction innovation within the technology space (Section 5.2); second, the Fire had a much stronger impact on non-wood (rather than wood) innovation and manufacturing (Section 6.2). Another approach to gauge this hypothesis is to contrast the case of Chicago with Boston, which also experienced a destructive fire in 1872.²⁸ While the Boston and Chicago fires differed in proportions, both caused massive damage and required extensive reconstruction efforts, resulting in large temporary economic booms (Miller, 2000; Hornbeck and Keniston, 2017). Impor-

²⁸An alternative explanation is that insurance companies in the aftermath of the Fire increased premiums for wooden buildings relative to non-wooden buildings, thereby incentivizing non-wood construction technologies. Since this response would plausibly have occurred in Boston as well as in Chicago, contrasting the reconstruction experiences of the two cities further informs the plausibility of this alternative explanation.

tantly, a key difference between the two contexts is that, while Chicago introduced comprehensive provisions prohibiting the use of wood and combustible materials in new buildings, this was not the case in Boston (see details in Section 2).

In Figure C.14, we display average plot-level land values in Chicago and Boston. Both fires targeted central parts of both cities, where land values, indicated by the red line, were on average higher than in the rest of the city, as shown by the gray line. In both cities, land values in areas exposed to the fire increased more than in unaffected areas. This comparison suggests that we can cautiously look at the Boston fire as a plausible counterfactual for the Chicago fire in the absence of construction policy interventions.

We begin by examining construction innovation in Boston following the 1872 Fire. Table VIII reports triple-differences estimates using Boston as the treated city. Columns (1–2) compare the number of patents in construction to other sectors in Boston and in control cities (excluding Chicago) before and after the 1872 Fire. We find a positive increase in construction patenting in Boston. On average over the sample period, construction patenting rises by approximately 19 percent relative to other sectors and cities, corresponding to about 9 additional patents (column 1). While this effect is statistically significant, its magnitude is roughly one-tenth of the corresponding estimate for Chicago. Moreover, the increase is concentrated in relatively low-novelty patents, as we detect no statistically significant response in high-novelty patenting following the Fire (column 2).

Columns (3–4) compare the responses of the number of patents in non-wood construction in Chicago to other cities and sectors. In contrast to Chicago, where non-wood construction innovation increased sharply after the Fire, we find a modest (11 percent) and statistically insignificant response in Boston. High-novelty non-wood construction innovation increases by 21%. Columns (5–6) compare wood construction to other sectors and find no evidence of increased innovation in this sector. Thus, although construction innovation increased in Boston, the gains in patenting activity are considerably smaller than in Chicago, especially in non-wood construction technologies. This contrast suggests that the anti-wood construction regulations enacted in Chicago played a central role in driving the shift toward non-wood construction innovation.

Panel B of Table III reports triple-difference estimates for construction manufacturing in Suffolk County, where Boston is located, relative to other sectors and counties. The results indicate a modest post-Fire expansion in construction activity. For example, the number of establishments increases by approximately 14 percent following the Fire. As in the patenting analysis, however, these effects are substantially smaller than those observed for Chicago. Taken together, the results indicate that while post-Fire reconstruction stimulated construction activity in both cities, the economic impact of the 1871 Chicago Fire was considerably larger. Historical evidence indicates that, in per capita terms, the monetary damages caused by the Boston Fire were approximately 44% of those in Chicago. Our estimates thus imply that economic growth ushered in by the Chicago Fire cannot be explained only by its size. Panel B of Table D.3 replicates a similar pattern when comparing Suffolk Counties with all other US counties, rather than only with those where the metropolitan areas in the sample are located.

Panel B of Table VI reports the response of non-wood construction manufacturing in Suffolk County

relative to other counties. Consistent with the null response of non-wood construction innovation to the 1872 Fire, we find no statistically significant effect of the Boston Fire on non-wood construction manufacturing across any of the outcomes considered. This pattern stands in sharp contrast to the large and positive effects documented for Chicago (Panel A), thus providing additional evidence that the post-Fire expansion of non-wood construction activity was specific to Chicago. Panel B of Table D.4 reports similar results when comparing Suffolk County with all other counties, although, in this case, we find an increase in non-wood manufacturing that is primarily driven by manufacturing in that sector in smaller counties experiencing a downward trajectory.

Finally, in columns (4–6) of Table VII, we estimate the difference-in-differences specification (2) using Boston, as opposed to Chicago, as the cross-sectional treatment variable (i.e., $I(c(i) = \text{Boston})$). We find that the probability of working in construction after the 1872 Fire in Boston increased by approximately 0.8%. This effect is roughly 45% as large as the increase in construction manufacturing we estimate in Chicago. In turn, neither non-wood (column 5) nor wood-related (column 6) construction employment displayed a statistically significant response to the 1872 Fire, unlike in Chicago. These results thus strongly suggest that the construction policy enacted by the Chicago municipal authority was key in shifting economic activity toward non-wood construction activity. These exercises indicate that the impact of the 1872 Great Boston Fire was, at best, modest compared to the 1871 Great Chicago Fire.

Altogether, while we cannot fully disentangle the effect of the fire from those of the policy, these results suggest that the construction policy enacted in Chicago in response to the 1871 fire might have played an important role in supporting innovation, manufacturing employment, and, ultimately, contributing to long-term economic growth.

7 Conclusions

Climate change is among civilization’s most pressing challenges. Worsening climate conditions are impacting the frequency and severity of natural disasters, which are expected to rise further over the coming decades. Adaptation and mitigation efforts critically hinge on technological change. Previous research documents that directed innovation can compensate for some of the adverse effects of natural disasters (Moscona and Sastry, 2023). In general, however, market competition may provide inefficiently low levels of resilience-enhancing innovation (Acemoglu, 2012).

This paper asks whether industrial policy can steer innovation to support technologies that mitigate the adverse effects of natural disasters. We study the 1871 Chicago Fire, which destroyed large parts of the city center. In response, the municipal authority forbade the construction of wooden buildings within the city perimeter.

Our analysis reveals that the 1871 Fire had large and positive effects on construction innovation in Chicago. The Fire fueled construction manufacturing across a broader set of economic activity indicators compiled from manufacturing censuses, such as the number of construction firms, their output, and various indicators of their size, and architectural heritage sites.

We employ two strategies to suggest that the municipal authority’s construction policy likely in-

fluenced the reconstruction and economic activity following the 1871 Fire. First, we analyze the heterogeneous treatment effects of the 1871 Fire on wood- and non-wood innovation and manufacturing in Chicago. Our results indicate that the Chicago Fire disproportionately fostered non-wood-related construction technical change and manufacturing firms, whereas the effects on wood construction are modest. Second, we study Boston in the aftermath of the 1872 Fire, which ravaged its business district but did not trigger any construction policy legislation. We find modest effects of the 1872 Fire on Boston’s construction innovation and manufacturing output.

Our results provide consistent evidence that Chicago’s construction policy contributed to channeling the construction demand shock generated by the 1871 Fire into non-wood construction, thereby propelling directed innovation. More generally, they reveal that public policy can sustain adaptation efforts and foster economic growth in response to ever-increasing concerns over deteriorating climatic conditions.

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Tables

Table I. Selected Descriptive Statistics

	Mean	Std. Dev.	Min	Max	Units	Obs.
	(1)	(2)	(3)	(4)	(5)	(6)
City-Level Descriptive Statistics						
Panel A. Innovation Activity						
Total Patents _{it}	1281.388	3299.580	0.000	37233.000	84	338
Construction Patents _{it}	217.905	672.832	0.000	7180.000	84	338
Wood Construction Patents _{it}	25.148	66.380	0.000	711.000	84	338
Non-Wood Construction Patents _{it}	48.112	151.456	0.000	1639.000	84	338
Panel B. Demographics						
Population _{it} (1,000s)	142.753	281.643	1.026	3377.385	84	336
Imputed Income per Capita _{it}	751.524	147.126	375.776	1113.352	84	336
Share of Men _{it} , %	49.610	2.429	40.941	67.933	84	336
Share Aged 0-25 _{it} , %	64.994	19.483	24.714	94.181	84	336
Share Aged 26-45 _{it} , %	23.289	11.286	5.250	53.604	84	336
Share Aged 46+ _{it} , %	11.717	8.595	0.483	31.532	84	336
Share Literate _{it} , %	61.708	13.870	22.098	81.187	84	336
Share of Non-White _{it} , %	10.901	19.388	0.049	86.644	84	336
Share of Foreign Born _{it} , %	20.313	12.351	0.091	52.767	84	336
Panel C. Employment Shares in Selected Occupations (%)						
Agriculture _{it}	6.717	6.505	0.158	46.092	84	336
Low-Skilled Manufacture _{it}	7.263	4.929	0.175	27.438	84	336
High-Skilled Manufacture _{it}	5.890	1.919	0.404	15.796	84	336
Laborer _{it}	4.881	1.854	0.095	14.989	84	336
Services _{it}	4.746	2.625	0.354	16.153	84	336
Panel D. Employment Shares in Selected Industries (%)						
Agriculture _{it}	6.967	6.457	0.309	46.152	84	336
Liberal Professions _{it}	7.434	2.974	1.540	19.142	84	336
Utilities _{it}	5.382	2.707	0.434	17.031	84	336
Construction _{it}	2.223	0.670	0.176	4.508	84	336
Textiles _{it}	1.840	3.473	0.000	20.760	84	336
County-Level Descriptive Statistics						
Panel E. Manufacturing Census						
N. of Establishments _{ct}	0.067	0.289	0.000	11.286	2087	6036
Production Value _{ct}	1.751	12.186	0.000	463.888	2087	6036
Fixed Capital _{ct}	0.918	5.683	0.000	186.674	2087	6036
Cost of Materials _{ct}	1.065	7.405	0.000	281.470	2087	6036
Cost of Labor _{ct}	0.315	2.304	0.000	95.319	2087	6036

Notes. This Table reports descriptive statistics for selected variables. Units are metropolitan areas in Panels A–D and counties in Panel E. All variables in Panels B–D are expressed as percentage shares of the city population—except for population and income per capita—and refer to censuses between 1850 and 1900. The number of establishments in Panel E is expressed in thousands of units; all other variables in Panel E are expressed in millions of dollars. Data in Panel E are tabulated from the Census of Manufacturing. Referenced on pages: [8](#), [9](#).

Table II. The Great Chicago Fire and Construction Innovation

	Construction		All Sectors	
	(1) All Patents	(2) Breakthrough Patents	(3) All Patents	(4) Breakthrough Patents
Chicago $\times$ Post-1871	1.108*** (0.066)	1.154*** (0.061)		
Chicago $\times$ Construction $\times$ Post-1871			0.226*** (0.049)	0.358*** (0.092)
City FE	Yes	Yes	–	–
Year FE	Yes	Yes	–	–
City-Year FE	–	–	Yes	Yes
City-Technology FE	–	–	Yes	Yes
Technology-Year FE	–	–	Yes	Yes
# Cities	83	79	83	83
# Observations	747	711	6,723	6,723
Mean Dep. Var.	93.008	19.316	49.053	10.228

Notes. This Table reports the effect of the Great Chicago Fire (1871) on construction innovation. In columns (1–2), the unit of observation is a city; in columns (3–4), the unit of observation is a city-technology pair. Units are observed at five-year intervals between 1855 and 1900. The dependent variable is the number of patents (columns 1 and 3) and the number of breakthrough patents (columns 2 and 4). In columns (1–2), the treatment is an interaction term equal to one for Chicago after 1871 and zero otherwise; in columns (3–4), the treatment is a triple interaction equal to one for Chicago and construction technology after 1871 and zero otherwise. All regressions are estimated through Poisson quasi-maximum likelihood. Regressions in columns (1–2) include city and year fixed effects; regressions in columns (3–4) include city-year, city-technology, and technology-year fixed effects. Standard errors are clustered at the city (columns 1–2) and city-technology (columns 3–4) levels. Referenced on page: 13. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

Table III. The Great Chicago and Boston Fires and Construction Manufacturing

	Number of Establishments	Production Value	Labor Costs	Material Costs	Capital
	(1)	(2)	(3)	(4)	(5)
Panel A: Cook County					
Cook County $\times$ Construction $\times$ Post-1871	0.320*** (0.076)	0.949*** (0.305)	0.668** (0.261)	1.047*** (0.285)	0.748** (0.291)
Mean Dep. Var.	4.373	13.806	12.425	13.063	13.192
Panel B: Suffolk County					
Suffolk County $\times$ Construction $\times$ Post-1872	0.132* (0.076)	0.754** (0.305)	0.543** (0.261)	0.601** (0.285)	0.447 (0.291)
Mean Dep. Var.	4.379	13.811	12.430	13.068	13.196
County-Year FE	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes
County-Industry FE	Yes	Yes	Yes	Yes	Yes
# Counties	75	75	75	75	75
# Observations	450	450	450	450	450

Notes. This Table reports the effect of the Great Chicago (1871) and Boston (1872) Fires on construction activity. The unit of observation is a county-industry pair observed at a decade frequency between 1860 and 1880. Industries are grouped into construction and other sectors. The dependent variable is: the number of establishments (column 1), production value (column 2), labor costs (column 3), material costs (column 4), and capital (column 5). All dependent variables are expressed in log terms. The treatment is an interaction term equal to one for the construction sector in Cook County (where Chicago is located, in Panel A) or Suffolk County (where Boston is located, in Panel B) in 1880, after their respective fires, and zero otherwise. All regressions are estimated through OLS. Regressions include county-year, county-industry, and industry-year fixed effects. Standard errors are clustered at the county-industry level. Referenced on pages: 16, 23. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

Table IV. The Great Chicago Fire and Landmark Sites

	Landmarks by Category:			All Categories
	(1) Pooled	(2) Architectural	(3) Non Architectural	(4)
Chicago $\times$ Post-1871	1.208*** (0.060)	1.379*** (0.060)	0.654*** (0.125)	
Chicago $\times$ Architecture $\times$ Post-1871				0.795*** (0.115)
City FE	Yes	Yes	Yes	–
Year FE	Yes	Yes	Yes	–
City-Year FE	–	–	–	Yes
City-Category FE	–	–	–	Yes
Category-Year FE	–	–	–	Yes
# Cities	84	84	74	84
# Observations	336	336	296	601
Mean Dep. Var.	10.244	9.158	1.233	5.727

Notes. This Table reports the effect of the Great Chicago Fire (1871) on landmark sites. In columns (1–3), the unit of observation is a city; in column (4), the unit of observation is a city-landmark category (architectural and non-architectural) pair. The sample in column (4) includes architectural and non-architectural landmarks. Units are observed at ten-year intervals between 1850 and 1900. The dependent variable is the number of landmarks (column 1), the number of architectural landmarks (column 2), and the number of non-architectural landmarks (column 3). In columns (1–3), the treatment is an interaction term equal to one for Chicago after 1871 and zero otherwise; in column (4), the treatment is a triple interaction equal to one for Chicago and architectural landmarks after 1871 and zero otherwise. All regressions are estimated through Poisson quasi-maximum likelihood. Regressions in columns (1–3) include city and year fixed effects; the regression in column (4) includes city-year, city-category, and category-year fixed effects. Standard errors are clustered at the city (columns 1–3) and city-category (column 4) levels. Referenced on page: 18. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

Table V. The Great Chicago Fire, Wood, and Non-Wood Construction Innovation

	Non-Wood Construction				Wood Construction			
	All Patents		Breakthrough Patents		All Patents		Breakthrough Patents	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Chicago $\times$ Post-1871	1.134*** (0.075)		1.361*** (0.084)		0.947*** (0.071)		0.792*** (0.107)	
Chicago $\times$ Non-Wood $\times$ Post-1871		0.181*** (0.066)		0.491*** (0.089)				
Chicago $\times$ Wood $\times$ Post-1871						-0.015 (0.078)		-0.073 (0.153)
City FE	Yes	–	Yes	–	Yes	–	Yes	–
Year FE	Yes	–	Yes	–	Yes	–	Yes	–
City-Year FE	–	Yes	–	Yes	–	Yes	–	Yes
City-Technology FE	–	Yes	–	Yes	–	Yes	–	Yes
Technology-Year FE	–	Yes	–	Yes	–	Yes	–	Yes
# Cities	79	83	76	82	81	83	74	82
# Observations	711	6,723	684	6,067	729	6,723	666	6,039
Mean Dep. Var.	21.556	40.998	4.846	9.617	10.951	39.906	2.045	9.338

Notes. This Table reports the effect of the Great Chicago Fire (1871) on non-wood and wood construction innovation. In columns (1), (3), (5), and (7), the unit of observation is a city; in columns (2), (4), (6), and (8), the unit of observation is a city-technology pair. In columns (1) and (3) (resp. 5 and 7), the sample includes non-wood (resp. wood) construction technologies; in columns (2) and (4) (resp. 6 and 8), the sample includes all technology classes except wood (resp. non-wood) patents. Construction patents that are neither wood- nor non-wood-related. Units are observed at five-year intervals between 1855 and 1900. The dependent variable is the number of patents (columns 1–2 and 5–6) and the number of breakthrough patents (columns 3–4 and 7–8). In columns (1), (3), (5), and (7), the treatment is an interaction term equal to one for Chicago after 1871 and zero otherwise; in columns (2) and (4) (resp. 6 and 7), the treatment is a triple interaction equal to one for Chicago and non-wood (resp. wood) construction technology after 1871 and zero otherwise. All regressions are estimated through Poisson quasi-maximum likelihood. Regressions in columns (1), (3), (5), and (7) include city and year fixed effects; regressions in columns (2), (4), (6), and (8) include city-year, city-technology, and technology-year fixed effects. Standard errors are clustered at the city (columns 1, 3, 5, and 7) and city-technology (columns 2, 4, 6, and 8) levels. Referenced on page: 19. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

Table VI. The Great Chicago and Boston Fires and Non-Wood Construction Manufacturing

	Number of Establishments	Production Value	Labor Costs	Material Costs	Capital
	(1)	(2)	(3)	(4)	(5)
Panel A: Cook County					
Cook County $\times$ Post-1871	1.133*** (0.104)	0.636* (0.324)	0.654** (0.301)	0.542* (0.292)	0.142 (0.312)
Mean Dep. Var.	2.649	11.108	10.103	10.065	10.531
Panel B: Suffolk County					
Suffolk County $\times$ Post-1872	0.028 (0.104)	-0.107 (0.324)	-0.394 (0.301)	-0.193 (0.292)	-0.188 (0.312)
Mean Dep. Var.	2.650	11.109	10.100	10.071	10.530
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
# Cities	73	73	73	73	73
# Observations	219	219	219	219	219

Notes. This Table reports the effect of the Great Chicago (1871) and Boston (1872) Fires on non-wood construction activity. The unit of observation is a county observed at a decade frequency between 1860 and 1880. The dependent variable is: the number of establishments (column 1), production value (column 2), labor costs (column 3), material costs (column 4), and capital (column 5). All dependent variables refer to the non-wood industries and are expressed in log terms. The treatment is an interaction term equal to one for Cook County (where Chicago is located, in Panel A) or Suffolk County (where Boston is located, in Panel B) in 1880, after their respective fires, and zero otherwise. All regressions are estimated through OLS. Regressions include county and year fixed effects. Standard errors are clustered at the county level. Referenced on pages: 20, 23. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

Table VII. The Great Chicago and Boston Fires and Construction Employment: Individual-Level Evidence

	Employed in:			Employed in:		
	(1) Construction	(2) Non-Wood Construction	(3) Wood Construction	(4) Construction	(5) Non-Wood Construction	(6) Wood Construction
Chicago $\times$ Post-1871	1.829** (0.716)	0.231*** (0.066)	0.604 (0.387)			
Boston $\times$ Post-1872				0.835* (0.446)	0.110 (0.138)	-0.010 (0.307)
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual Controls	Yes	Yes	Yes	Yes	Yes	Yes
# Cities	84	84	84	84	84	84
# Observations	1,974,963	1,974,963	1,974,963	1,977,763	1,977,763	1,977,763
Mean Dep. Var.	1.592	0.255	1.044	1.585	0.254	1.039

Notes. This Table reports the effect of living in Chicago (columns 1–3) and Boston (4–6) on subsequent employment in construction. The unit of observation is an individual, observed two times in the 1870 and 1880 population censuses. The dependent variable is a dummy equal to 100 if the individual in 1880 is recorded working in construction (columns 1 and 4), non-wood-related construction (columns 2 and 5), and wood-related construction (columns 3 and 6). The sample includes individuals who were not working in construction in 1870. The treatment is an interaction term between an indicator variable equal to one if the individual is recorded as living in Chicago (columns 1–3) or Boston (columns 4–6) and zero otherwise, and a dummy equal to one for the post-Fire observation (1880) and zero otherwise. Each specification includes city and census year fixed effects and further controls for individual cohort, race, literacy, and internal migration. Standard errors are clustered at the metropolitan area level. Referenced on pages: 22, 24. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

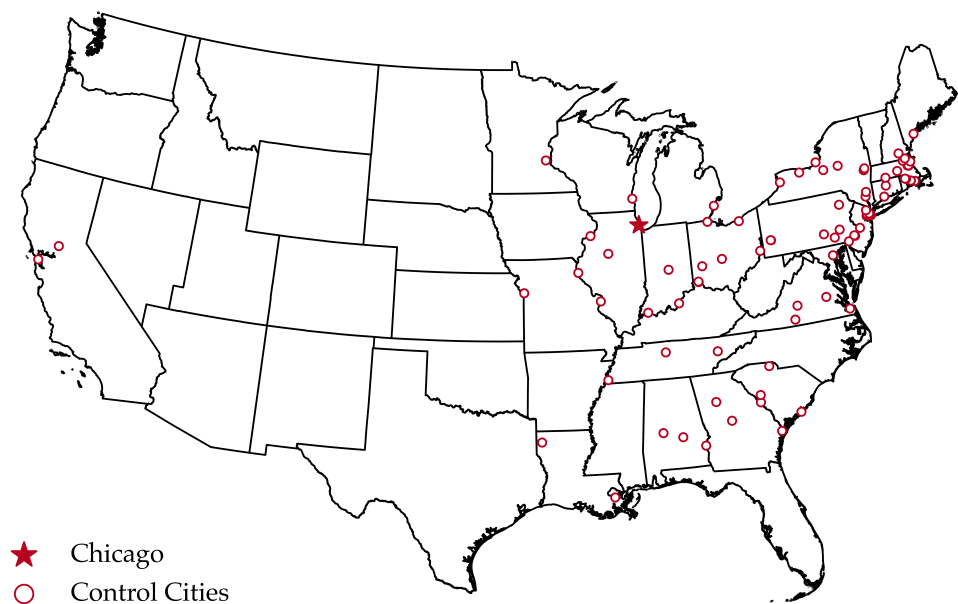
Table VIII. The Great Boston Fire, Wood, and Non-Wood Construction Innovation

	Construction		Non-Wood Construction		Wood Construction	
	(1) All Patents	(2) Breakthrough Patents	(3) All Patents	(4) Breakthrough Patents	(5) All Patents	(6) Breakthrough Patents
Boston $\times$ Construction $\times$ Post-1872	0.178*** (0.067)	0.016 (0.063)				
Boston $\times$ Non-Wood Construction $\times$ Post-1872			0.118 (0.104)	0.199** (0.086)		
Boston $\times$ Wood Construction $\times$ Post-1872					-0.173* (0.094)	-0.159 (0.135)
City-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
City-Technology FE	Yes	Yes	Yes	Yes	Yes	Yes
Technology-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
# Cities	83	83	83	82	83	82
# Observations	6,450	6,723	6,723	6,067	6,723	6,039
Mean Dep. Var.	45.198	9.954	37.209	9.427	36.215	9.182

Notes. This Table reports the effect of the Great Boston Fire (1872) on construction innovation. The unit of observation is a city-technology pair. Units are observed at five-year intervals between 1855 and 1900. The dependent variable is the number of patents (columns 1, 3, and 5) and the number of breakthrough patents (columns 2, 4, and 6). In columns (1–2), the treatment is an interaction term equal to one for the construction technology in Boston after 1872 and zero otherwise; in columns (3–4) (resp. 5–6), the treatment is a triple interaction equal to one for Boston and non-wood (resp. wood) construction technology after 1872 and zero otherwise. The sample in columns (1–2) includes all technologies excluding wood- and non-wood construction technologies, which are included in the construction technology category; the sample in columns (3–4) (resp. 5–6) excludes wood (resp. non-wood) construction technologies and construction patents that are neither wood- nor non-wood-related. All regressions are estimated through Poisson quasi-maximum likelihood. All regressions include city-year, city-technology, and technology-year fixed effects. Standard errors are clustered at the city-technology level. Referenced on page: 23. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

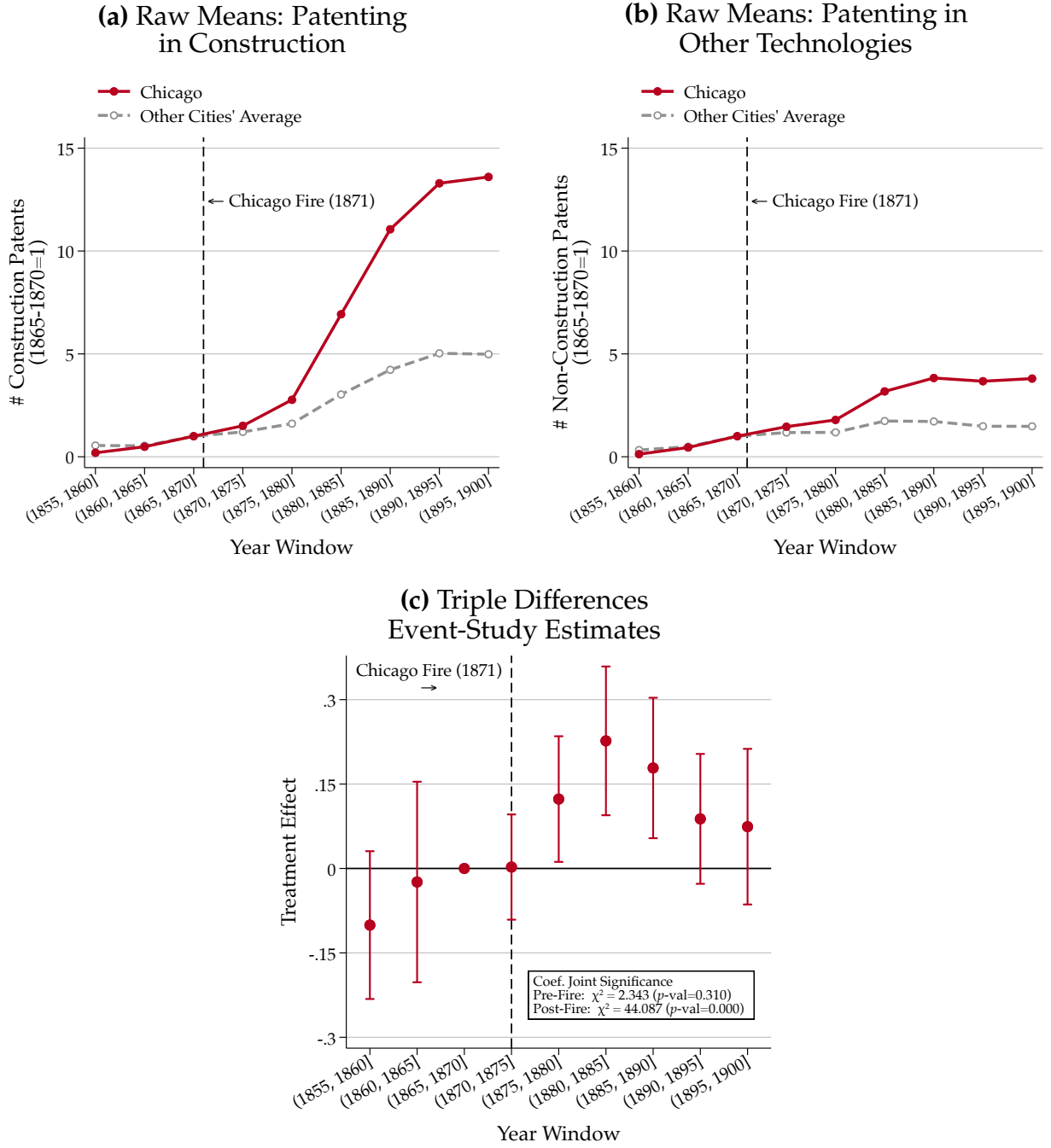
Figures

Figure I. Metropolitan Areas Above 20,000 Population



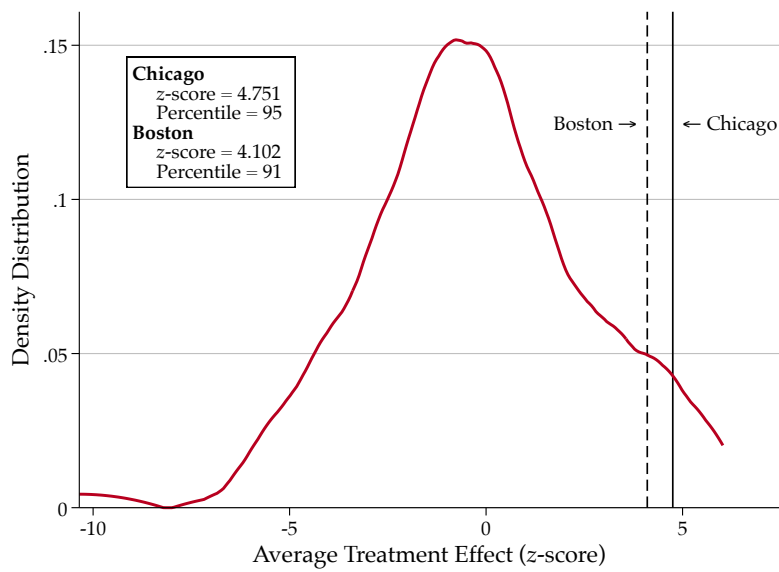
Notes. This map reports the location of the 84 metropolitan areas in the analysis sample. The coordinates of each metropolitan area report the center of its largest city. To construct the metropolitan areas, we retain all cities above 20,000 inhabitants; then, we agglutinate each minor city below the threshold to its closest city within 20 kilometers above the 20,000 population threshold. The red star reports the location of Chicago; the red dots report the location of all other cities. Cities are overlaid on state borders in 1870, before the Great Chicago Fire (1871). The full list of metropolitan areas is reported in Appendix Table D.2. Referenced on page: 10.

Figure II. Innovation in Construction in Chicago



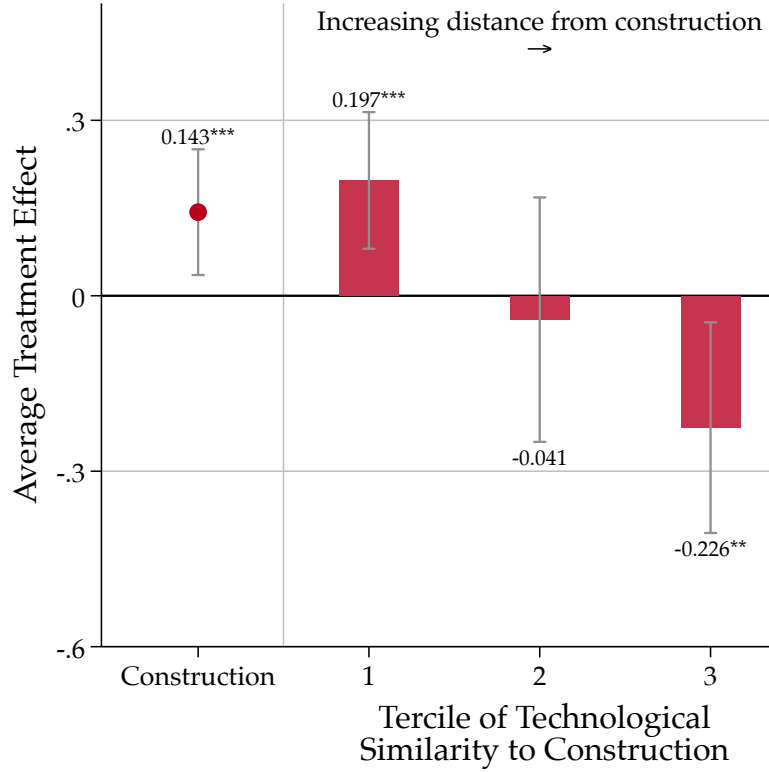
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on construction innovation in Chicago. In Panel IIa, we report the number of patents in construction in Chicago (red) and in other cities (gray), between 1855 and 1900. Panel IIb replicates the graph for patents in all sectors except construction. The number of patents issued in the 1865–1870 five-year window, i.e., the last one before the 1871 Fire, is normalized to one. In Panel IIc, we report the triple-differences dynamic treatment effects estimates. The unit of observation is a city-technology pair, and units are observed over five-year windows between 1855 and 1900. The dependent variable is the number of patents issued in each year window. Each dot reports the coefficient for an interaction between period dummies and a double interaction term equal to one for Chicago and the construction technology class, and zero otherwise. The model is estimated by Poisson quasi maximum likelihood, and it includes city-year, city-technology, and technology-year fixed effects. Standard errors are clustered at the city-technology level; bands report 95% confidence intervals. Referenced on page: 13.

Figure III. Randomization Test of Innovation in Construction



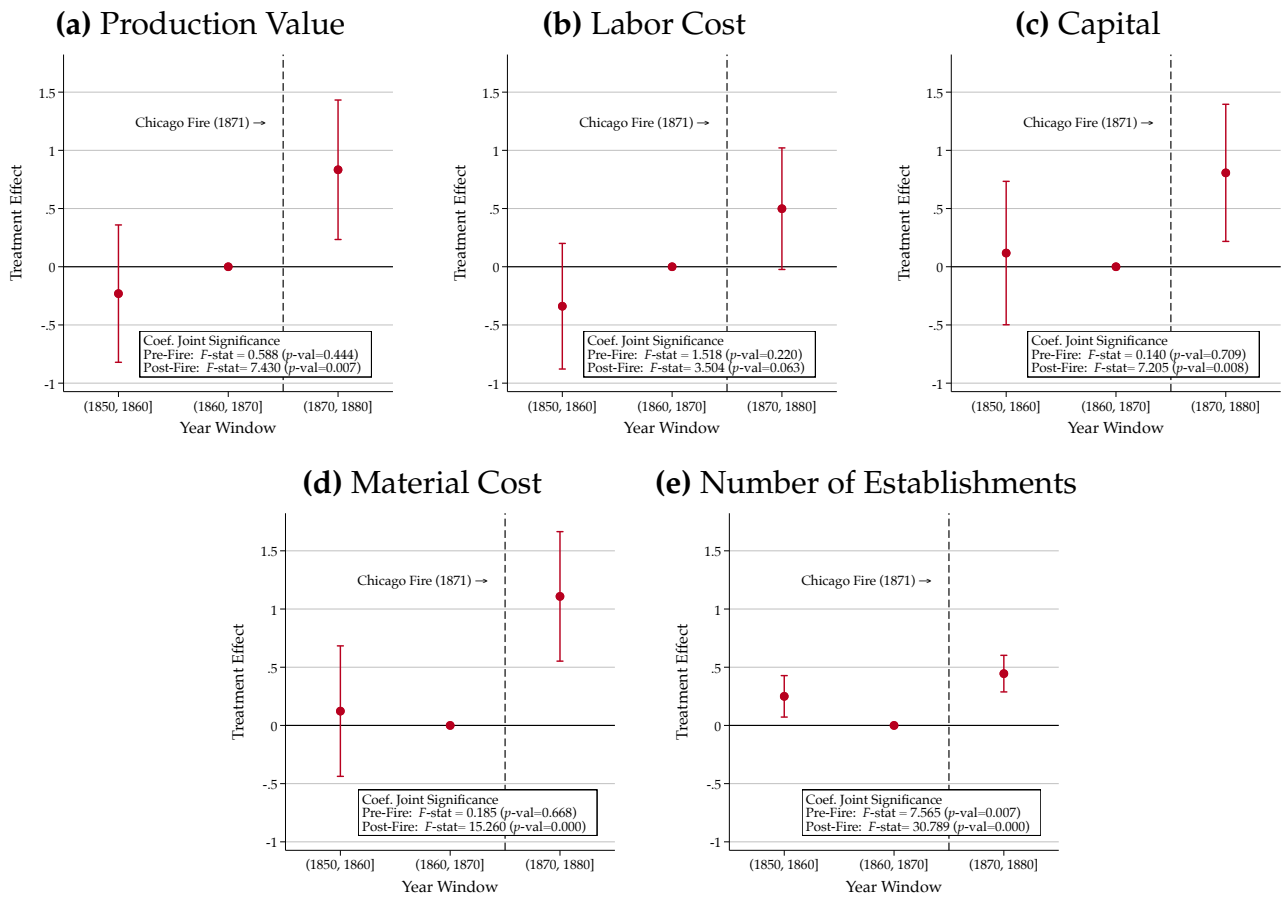
Notes. This Figure reports a randomization test on the effect of the Great Chicago Fire (1871) on construction innovation. The unit of observation is a city-technology pair, and units are observed over five-year windows between 1855 and 1900. The dependent variable is the number of patents issued in each year window. For each city, we estimate the average treatment effect as the coefficient of an interaction term equal to one for patents issued after the Chicago Fire in construction in the given city. The red line reports the distribution of the average treatment effects thus estimated. The black solid and dashed lines report the average treatment effect obtained when the treated cities are Chicago and Boston, respectively. The model is estimated by Poisson quasi maximum likelihood, and it includes city-year, city-technology, and technology-year fixed effects, as well as city-technology time trends. The Figure reports the z-score of the average treatment effect for Chicago and Boston, as well as their percentile over the distribution of average treatment effects. Referenced on page: [14](#).

Figure IV. Spillover Effects of the Great Chicago Fire on Innovation



Notes. This Figure reports the effect of the Great Chicago Fire (1871) on construction and non-construction innovation. The unit of observation is a city-technology pair, and units are observed over five-year windows between 1855 and 1900. The dependent variable is the number of patents issued in each year window. Each bar reports the coefficient for a triple interaction term equal to one for patents issued in Chicago after 1871 in technology classes within the k -tercile of similarity with construction and zero otherwise. Specifically, for each similarity tercile with construction ξ_k , we estimate $\ln \mathbb{E} [\text{Patents}_{ikt}] = \alpha_{ik} + \alpha_{it} + \alpha_{kt} + \beta \cdot I(i = \text{Chicago}) \cdot I(k \in \xi_k) \cdot I(t \geq 1871)$, where i , k , and t denote cities, technology classes, and time periods. Technologies in the first tercile ($\xi_k = 1$) are the most similar to construction; technologies in the last tercile ($\xi_k = 3$) are the least similar to construction. The sample excludes construction patents. The red dot reports the average treatment effect on construction. All regressions are estimated by Poisson quasi-maximum likelihood and include city-year, city-technology, and technology-year fixed effects. Standard errors are clustered at the city-technology level; bands report 95% confidence intervals. Referenced on pages: 15, 16. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

Figure V. Dynamic Effect of the Great Chicago Fire on Construction Manufacturing Activity



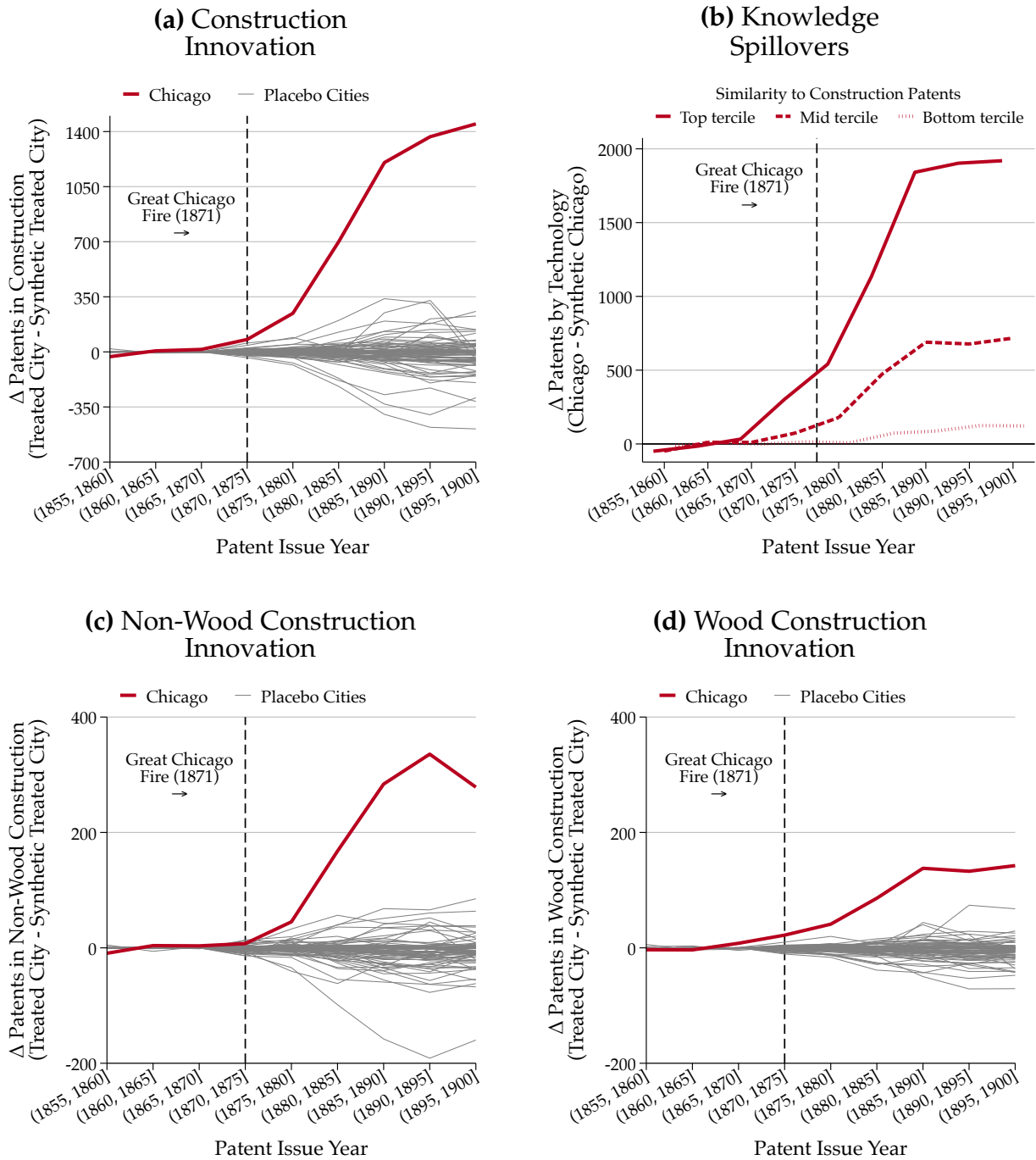
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on manufacturing activity. The unit of observation is a county-industry pair observed at a decade frequency in 1860, 1870, and 1880. Industries are grouped into construction and non-construction industries. The dependent variable is: production value (Panel Va), labor cost (Panel Vb), capital (Panel Vc), material cost (Panel Vd), number of establishments (Panel Ve). Each dependent variable is expressed in log terms. Each dot reports the coefficient of an interaction between decade dummies and a double interaction equal to one for construction industries in Cook County—where Chicago is located—and zero otherwise. The model is estimated by OLS, and it includes county-year, county-industry, and industry-year fixed effects. Standard errors are clustered at the county-industry level; bands report 95% confidence intervals. The 1870 census year serves as the baseline period. Each panel reports the results of tests for the joint significance of pre- and post-treatment coefficients. Referenced on page: 16.

Figure VI. Manufacturing Spillovers of the Great Chicago Fire



Notes. This Figure reports the effect of the Great Chicago Fire (1871) on the production value of manufacturing firms by industry. The unit of observation is a county-industry pair observed at a decade frequency in 1860, 1870, and 1880. The dependent variable is the log-production value. For each industry, we estimate a separate regression and report the coefficient of a triple interaction equal to one for that industry in Cook County—where Chicago is located—after 1871 and zero for all other industries and counties. The model is estimated by OLS, and it includes county-year, county-industry, and industry-year fixed effects. Standard errors are clustered at the county-industry level; bands report 95% confidence intervals. We color in red the construction sector. Referenced on page: [16](#).

Figure VII. Construction Innovation in Chicago: Synthetic Control Estimates



Notes. This Figure reports the effect of the Great Chicago Fire (1871) on construction innovation using a synthetic control framework. The unit of observation is a metropolitan area at a five-year frequency between 1855 and 1900. The dependent variable is: in VIIa, the number of patents in construction; in VIIb, the number of patents by tercile of technology class-similarity to construction patents; in VIIc and VIId, the number of patents in construction that are not related to the use of wooden materials and those that are related to wooden materials. In each graph except Panel VIIb, the red line reports the difference between Chicago and synthetic Chicago, and the gray line reports the difference between other cities and their synthetic counterpart. In Panel VIIb, each line reports the difference between Chicago and synthetic Chicago: the solid line reports the effect on the top tercile of most similar technology classes to construction, the dashed line reports the effect on the mid tercile, and the dotted line reports the effect on the bottom tercile. In each graph, the black dashed line reports the date of the Great Chicago Fire (1871). Referenced on pages: 14, 16, 20.

ONLINE APPENDIX

Natural Disasters, Industrial Policy, and Innovation

Davide M. COLUCCIA & Mara P. SQUICCIARINI

May, 2026

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A Data Appendix

This section provides further information on the primary data sources underlying the datasets used in the main analysis and the methods adopted to construct them.

A.1 Patent Data

We collect a novel dataset of US patents spanning 1855–1900. Compared to existing datasets, we develop a new methodology to extract information from patent texts. This section describes this new repository.

A.1.1 *Motivation*

Our analysis requires detailed information on each inventor’s location to assign the patents to the closest metropolitan area. We also need text data to identify construction, wood- and non-wood-related innovations, and technological classes. Additionally, the United States Patent Office number allows for matching patent records to the novelty measure produced by Kelly et al. (2021). We now briefly discuss why publicly available datasets do not satisfy these data requirements.

There are three datasets of US patents covering the period 1850–1900 (Andrews, 2021). Among those, the data produced by Sarada et al. (2019) and Petralia et al. (2016) are publicly available. Sarada et al. (2019) digitize the Annual Reports of the Commissioner of Patents and the Annual Indices of Patents. This dataset starts in 1870 and does not contain geographical information on inventors’ addresses beyond their state. In addition, it does not include text data, as the primary source is not the text of the patent but an index.

Petralia et al. (2016) extracted patent information from digitized images of US patents, as we do. The data contain information on the county of the inventors. This data is, therefore, not suited for our analysis, which is performed at a more granular level. Importantly, this dataset does not contain the text of the patents, which we use to identify construction and wood- and non-wood-related construction innovations.

A.1.2 *Methodology to Assemble the Dataset*

We rely on the corpus of digitized US patents produced by Google Patents. We collect the universe of patent documents by adapting the algorithm developed by Moser and San (2020). On top of the patent images, we collect from Google Patents the CPC technological classes of each patent and the backward and forward citations, which we do not use in this paper.

From the patent texts, we follow a procedure similar to Coluccia and Dossi (2025), who apply it to historical British patents. We use a large language model—GPT 4o-mini—to extract the name and surname of the inventors, their address of residence, the filing and issue date of the patent, and information on whether the patent has a firm assignee. Compared to standard extraction methods relying on regular expressions, the flexibility of large language models allows us to parse the data even in the presence of minor OCR errors or syntactical inconsistencies. We employ the following prompt.

gpt-4o-mini Prompt for Patent Information Extraction

The following is the text of a US patent.

In a JSON object, for each inventor, report:

- Full name,
- Location,
- Publication date,
- Filing date,
- Assignee,
- Assignee location.

Code missing information as “N/A”.

Only return inventors’ records when they are clearly listed in the patent’s text. The patent may not contain all the required information.

Text of US Patent: < Text of Patent >

Lastly, we use a commercial georeference software—Google Maps API—to assign latitude and longitude coordinates to the address of each inventor. The address listed on the patent is the residence of the inventor. In small towns, the address indicates the town. In larger cities, the proper address appears on the patent. For consistency, we georeference the town, given that the analysis is performed at that level of spatial aggregation.

Following standard practice in the literature, we date patents by the inventor’s filing year. In the early part of the sample, filing dates are missing for a subset of patents. We impute these filing years by subtracting the average delay between issue and filing year from the recorded issue year. The average delay is computed on the subset of patents that report both issue and filing years. Approximately 89% of patents report both the issue and the filing year, implying that we impute the filing year for the remaining 11%.

A.1.3 Comparison with Existing Datasets

The dataset contains information on all patents listed on Google Patents issued between 1855 and 1900. We have the full digitized text of the patent, the name and surname of the inventor(s), their address(es), their latitude and longitude, the filing and issue date of the patent, and an assignee flag. Since the dataset contains the unique patent number, we can match our records with the novelty measure assembled by Kelly et al. (2021). From Google Patents, we also have the CPC technological classification associated with each patent.

Compared to Andrews (2021), whose coverage spans 1870 to 1942, our data has more detailed geographical information, more systematic inventors’ names and surnames, full-text access, CPC technology classification information, filing and issue years, and more precise dates, down to the day.

Compared to Petralia et al. (2016), our dataset contains more granular geographical information—down to the georeferenced town of each inventor, as opposed to the county—, CPC classes, filing and issue dates, and full-text access.

A.1.4 External Validation

Even though the existing datasets are not suited for our empirical application, we use them to validate the coverage of our new sample. Figure C.1a compares the number of patents in our dataset with Petralia et al. (2016)—dashed gray line—and Sarada et al. (2019)—dotted gray line. Figure C.1b reports coverage rate, defined as the number of patents in our dataset relative to Petralia et al. (2016)—red solid line—and Sarada et al. (2019)—gray dashed line, expressed in percentage.

Our series mimics Petralia et al. (2016) throughout the period. The coverage rates vary between 90% before the 1870s and 110% during the later part of the period, indicating that our dataset contains more patents. The data by Sarada et al. (2019) is less comprehensive—our dataset contains one-third more patents between 1870 and 1880 and one-tenth more towards the end of the century—but displays a similar co-movement with our series. Importantly, even though we cannot perform validation with the dataset developed by Berkes (2018), their data and that of Petralia et al. (2016) display substantial overlap before 1900, which indicates that our coverage rate of his dataset will mimic that with Petralia et al. (2016).

A.1.5 Methodology to Identify Construction Patents

We adopt a simple dictionary-based procedure to identify patents related to construction, non-wood construction, and wood construction.

First, we generate a dictionary of 30 words related to construction using GPT 5. The words are displayed in Table D.1. Then, we search for the number of instances each word appears in the text of each patent. A patent is identified as “construction-related” if at least one word appears at least five times in its text. The results are not sensitive to alternative, more demanding thresholds, but increasing the required threshold increases the rate of false negatives without significantly impacting the rate of false positives. We provide a more detailed discussion below.

Second, *among construction patents*, we search for wood- and non-wood-related innovations. Using GPT 5, we generate a list of 12 wood-related and 11 non-wood-related words. We then count the number of instances each word appears in the patents’ texts. A patent is identified as “wood-related construction” if (i) it is a construction patent, and (ii) it mentions at least word from those in from column (2) of Table D.1. Analogously, a patent is identified as “non-wood-related construction” if (i) it is a construction patent, and (ii) it mentions one or more words from column (3) of Table D.1.

We manually check the plausibility of the results obtained using this dictionary-based approach on a random sample of 200 patents. Within this sample, 41 patents were flagged as construction-related, five as wood-construction, and six as non-wood-construction. We did not find false positives, and the algorithm missed four patents that a human would have coded as construction, thus yielding a 10% false negatives rate among construction patents.

A.2 Population Census

We use individual-level data mapped to CPP locations to compute demographic characteristics of the metropolitan areas and counties. The synthetic control and synthetic difference-in-differences estimates use, as balancing variables, the share of natives, blacks, imputed income per capita (OCCSCORE), and the employment shares by occupation and industry.

To compute the employment share by occupation, we use the OCC1950 standardized codes to construct a coarser occupational taxonomy that follows from the categories provided by IPUMS: professionals (OCC1950 between 0 and 99), farmers (OCC1950 between 100 and 123 and 810 and 840), managers (OCC1950 between 200 and 290), clerical and sales (OCC1950 between 300 and 390), craftsmen (OCC1950 between 500 and 595), operatives (OCC1950 between 600 and 690), services (OCC1950 between 700 and 790), and laborers (OCC1950 between 910 and 970).

Similarly, to construct the employment share by industry, we use the IND1950 standardizes codes to construct a coarser industry taxonomy following IPUMS: agriculture (IND1950 between 105 and 126), chemicals (IND1950 between 466 and 478), construction (IND1950 between 246 and 246), engineering (IND1950 between 367 and 388 or 898), liberal professions (IND1950 between 716 and 897), metallurgy (IND1950 between 336 and 348), miscellaneous manufacturing (IND1950 between 306 and 358, 406 and 429, and 456 and 459), public administration (IND1950 between 906 and 946), textiles (IND1950 between 436 and 449), trade (IND1950 between 606 and 699), transportation (IND1950 between 506 and 579), and utilities (IND1950 between 596 and 598 and 826 and 859).

All variables from the population census are measured in 1870, i.e., the year before the Great Chicago Fire.

A.3 Manufacturing Census

We compile manufacturing data—number of establishments, value of production, labor cost, material cost, and capital—from county-by-industry data from the Census of Manufactures transcribed by Hornbeck and Rotemberg (2024). The sample comprises all counties with at least one metropolitan area.

We use the industry concordance table provided by the authors to construct a county-by-industry panel at the decade level between 1860 and 1880. The final sample is a decadal county-level panel that records the various variables for different industries. In particular, we group the “construction,” “construction materials,” and “furniture” as one “construction” industry, the “brick, stone, and tile,” “marble and stone work,” and “lime and cement” titles as one “non-wood manufacturing” industry, and the “carpentering” title as one “wood manufacturing” industry. This categorization follows the taxonomy produced by Hornbeck and Rotemberg (2024). Results remain unchanged if we include “lumber, sawed,” “lumber, planed,” “wood products, other,” and “wood, turned and carved” as plausible wood-related construction industries.

A.4 Historical Landmarks

We collect all historical landmarks from the “National Register of Historic Places” (Stutts, [2024](#)). We georeference each entry using the provided address and Google Maps API. We assign latitude and longitude to 99% of the landmarks in the sample. Each landmark is then assigned to the closest CPP location within 20 km, following the same procedure we use for the patent records. The National Register does not contain information on the construction year. We thus manually search each entry on Wikipedia and assign a construction year to 77% of the entire dataset. Among the landmarks in the metropolitan analysis sample, 85% have a recorded construction year. We include in the sample entries that refer to “buildings” (64% of the sample), thus excluding “districts” (20%), “objects” (0.25%), and “sites” (13%).

B Summary of the Robustness Analyses

This section provides complementary information on the robustness exercises mentioned in passing in the main text.

B.1 Patent Novelty Data

We use the text-based measure by Kelly et al. (2021) to explore the effect of the 1871 Fire on economically relevant innovation. Intuitively, a patent that is more similar to future patents than to previous patents is labeled as “more innovative.” More formally, let the backward inverse-document frequency associated with word w be defined as

$$\text{BIDF}_{w,t} \equiv \log \left(\frac{\text{Number of Patents Issued Before } t}{1 + \text{Number of Patents Issued Before } t \text{ that contain word } w} \right). \quad (\text{B.1})$$

To each patent-word pair, it associates a variable equal to the number of instances word w appears in patent i , normalized by the patent length. Let i denote the patent and the set of words it contains. The term-frequency weight is equal to

$$\text{TF}_{wi} \equiv \frac{\sum_{c \in i} 1(c = w)}{\sum_{c \in i} 1(c)}, \quad (\text{B.2})$$

where the numerator returns how many times word w appears in patent i , and the denominator is the number of words in patent i . The TF-BIDF associated with word w , patent i at time t is the product between the TF and the BIDF:

$$\text{TF-BIDF}_{wi,t} \equiv \text{TF}_{wi} \times \text{BIDF}_{w,t}. \quad (\text{B.3})$$

The vector $\text{TF-BIDF}_{i,t}$ thus collects the $\text{TF-BIDF}_{wi,t}$ for all words w in i , normalized by its norm to have unit length.

The approach allows the representation of each patent as a TF-BIDF vector. One can thus compute a measure of similarity—in their case, the cosine similarity—between each patent pair. In particular, the backward similarity is the average similarity between i and all previous patents within τ_1 years:

$$\text{Backward Similarity}_i^{\tau_1} \equiv \frac{1}{|\mathcal{F}_i^{-\tau_1}|} \sum_{j \in \mathcal{F}_i^{-\tau_1}} \rho_{i,j}, \quad (\text{B.4})$$

where the set $\mathcal{F}_i^{-\tau_1}$ denotes the set of US patents issued within τ_1 years from the issue year of patent i and $\rho_{i,j}$ is the cosine similarity between the vectors $\text{TF-BIDF}_{i,t}$ and $\text{TF-BIDF}_{j,t}$. Analogously, the forward similarity is the average similarity with all patents in the later τ years:

$$\text{Forward Similarity}_i^{\tau_2} \equiv \frac{1}{|\mathcal{F}_i^{+\tau_2}|} \sum_{j \in \mathcal{F}_i^{+\tau_2}} \rho_{i,j}, \quad (\text{B.5})$$

where the set $\mathcal{F}_i^{+\tau_2}$ denotes the set of US patents issued τ_2 years after the issue year of patent i . Given these measures, one can compute the similarity of i with future relative to previous patents:

$$\text{Excess Forward Similarity}_i^{\tau_1, \tau_2} \equiv \frac{\sum_{j \in \mathcal{F}_i^{+\tau_2}} \rho_{i,j}}{\sum_{j \in \mathcal{F}_i^{-\tau_1}} \rho_{i,j}}. \quad (\text{B.6})$$

In our application, we take $\tau_1 = \tau_2 = 5$ to have a symmetric 5-year window around each patent. The

results remain qualitatively unchanged when using 1- and 10-year symmetric windows. Following Kelly et al. (2021), we partial out year fixed effects from the raw excess similarity measure to ensure that aggregate trends in language and patent redaction do not influence the results. A patent is then defined as “novel” if it is in the top 20% of the excess forward similarity distribution. The results remain qualitatively unchanged using different thresholds at the 5% and 10%.

B.2 Synthetic Difference-in-Differences

The synthetic difference-in-differences estimator developed by Arkhangelsky et al. (2021) nests the insights of standard synthetic control and difference-in-differences estimators. As highlighted by Arkhangelsky et al. (2021), the synthetic control method is typically applied in settings with one or a few treated units, where the parallel trends assumption required by the difference-in-differences estimator is unlikely to hold. The synthetic difference-in-differences estimator weights units in the control group to match the treated unit in terms of a set of specified pre-treatment observable characteristics and the outcome to maximize the empirical plausibility of the parallel trends assumption.

Formally, let i and t denote units and time periods. Exposure to the treatment is $W_{it} = \{0, 1\}$, and Y_{it} denotes the outcome. the synthetic control estimator selects weights ω_i^{sc} to minimize the distance between treated and control units and estimates the treatment effect as

$$(\hat{\tau}^{sc}, \hat{\mu}, \hat{\beta}) = \arg \min_{\mu, \beta, \tau} \left\{ \sum_i \sum_t (Y_{it} - \mu - \beta_t - W_{it}\tau)^2 \hat{\omega}_i^{sc} \right\}, \quad (\text{B.7})$$

where $\hat{\tau}^{sc}$ is the estimated treatment effect. The difference in differences estimator, by contrast, weights all units in the same way, but it includes unit fixed effects to leverage within-unit variation:

$$(\hat{\tau}^{did}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\mu, \alpha, \beta, \tau} \left\{ \sum_i \sum_t (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \right\}. \quad (\text{B.8})$$

The synthetic difference-in-differences estimator nests these two approaches. First, it selects weights ω_i^{sdi} to minimize the distance between treated and control units in terms of pre-treatment outcome values and characteristics. Moreover, it selects λ_t^{did} that balance pre-exposure time periods with post-exposure ones. Then, it solves for the average treatment effect as in the DiD estimator, applying the so-computed weights:

$$(\hat{\tau}^{sdi}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\mu, \alpha, \beta, \tau} \left\{ \sum_i \sum_t (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \hat{\omega}_i^{sdi} \hat{\lambda}_t^{did} \right\}. \quad (\text{B.9})$$

By weighting observations to minimize the distance between treated and control units, the SDiD estimator emphasizes units that are, on average, similar to the treated unit and periods that are, on average, similar to the target periods.

In our application, we compute the SDiD weights by including the population, share of whites and foreign-born, employment shares by occupation, and employment shares by industry. As in the synthetic control baseline case, these variables were measured in the 1870 census, before the Chicago and Boston fires.

We display the SDiD results in terms of standard panel event-study estimates to visualize the evo-

lution of treatment effects over time. In addition, by looking at the pre-treatment differences between the treated (Chicago and Boston) and control units, we can assess the empirical plausibility of the parallel trends assumption. We follow the logic outlined in Clarke, Paila  ir, Athey and Imbens (2023) to construct these figures. We wish to estimate, for each period t ,

$$(\bar{Y}_t^T - \bar{Y}_t^C) - (\bar{Y}_0^T - \bar{Y}_0^C), \quad (\text{B.10})$$

where $\bar{Y}_t^T$ and $\bar{Y}_t^C$ denote the average outcome for treated and control units at time t , and $\bar{Y}_0^T$ and $\bar{Y}_0^C$ denote the average pre-treatment outcome values for treated and control units. These are computed as

$$\bar{Y}_0^T = \sum_{t=1}^{\tau-1} \hat{\lambda}_t^{sdiid} \bar{Y}_t^T, \quad (\text{B.11})$$

for the treatment group and, similarly, as

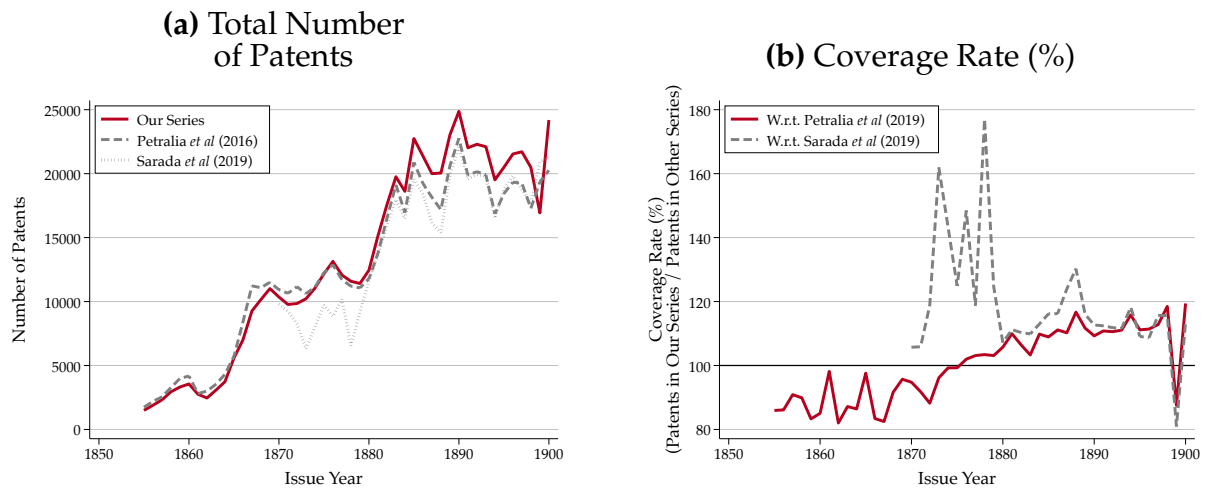
$$\bar{Y}_0^C = \sum_{t=1}^{\tau-1} \hat{\lambda}_t^{sdiid} \bar{Y}_t^C, \quad (\text{B.12})$$

for the control group. The term τ denotes the treatment period. To compute the confidence intervals, we apply block-bootstrap resamples with 100 replications.

We display SDiD estimates for all the results shown in the main text in Figure C.5. In all cases, the synthetic DiD estimates confirm the baseline results obtained through the more traditional synthetic control approach. Additionally, in most cases, the pre-treatment differences between treated and control units are statistically insignificant and are always quantitatively very small. These patterns confirm the empirical plausibility of the parallel trends assumption that requires that, in the absence of the Fires, the outcomes in Chicago and Boston and in the control units would have evolved similarly.

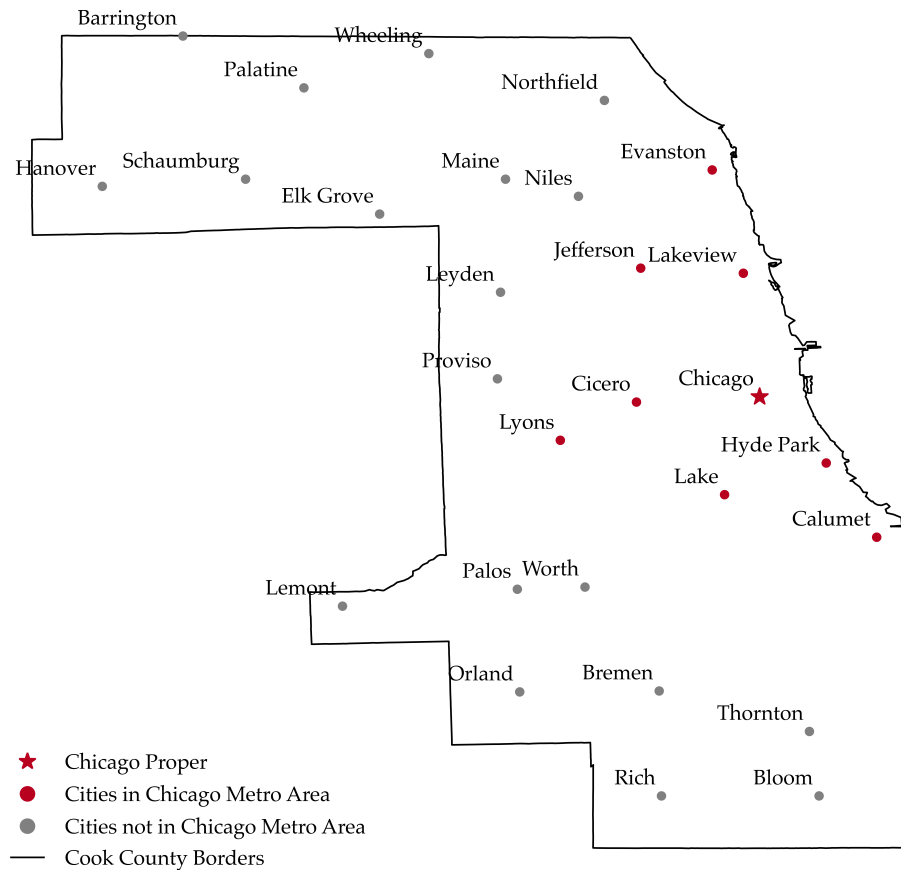
C Additional Figures

Figure C.1. Comparison of Own and External Patent Repositories



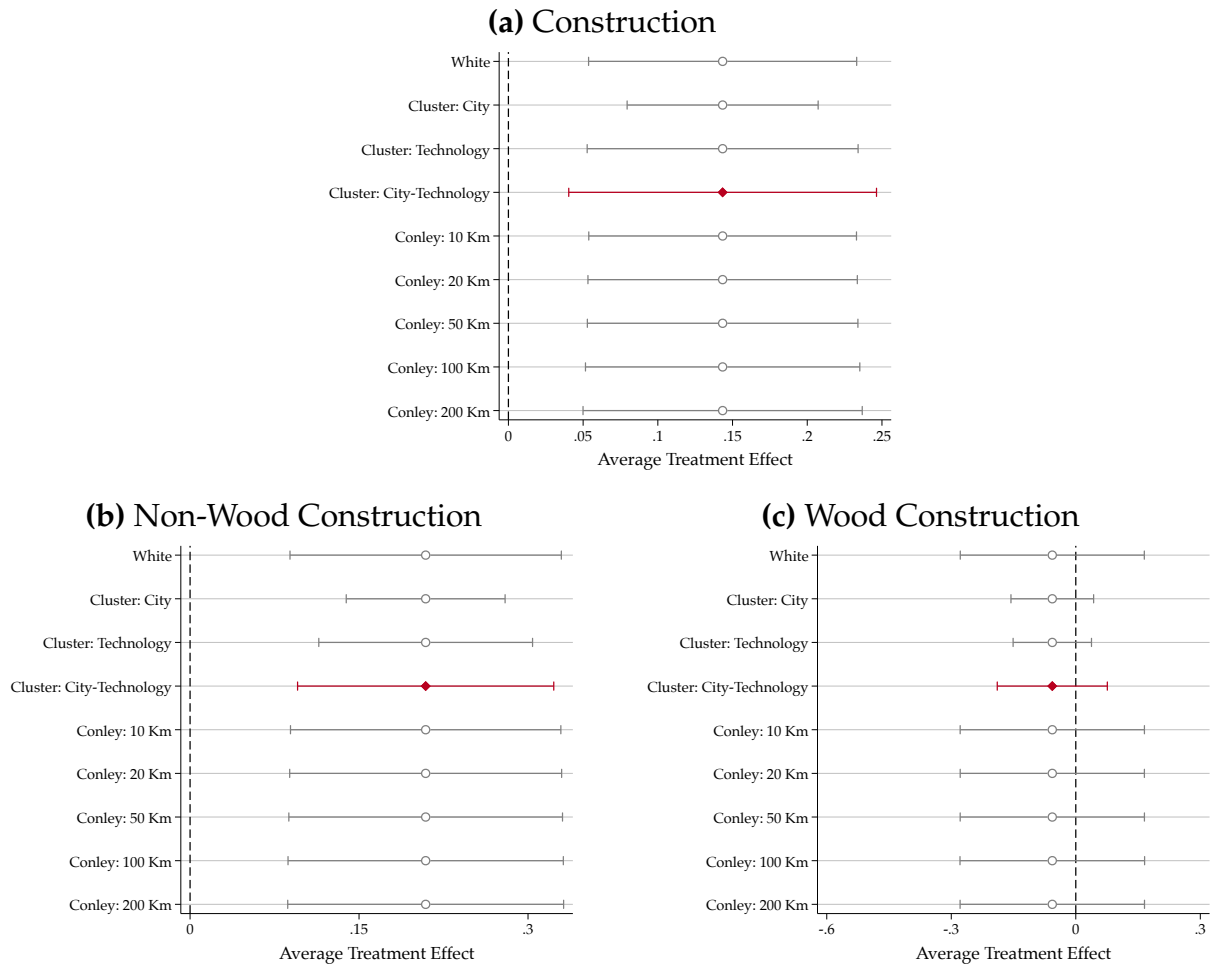
Notes. This Figure compares the number of patents in the dataset produced for this paper and data from Petrulia et al. (2016) and Sarada et al. (2019). Panel C.1a reports the total number of patents in our data (red line), Petrulia et al. (2016) (gray dashed line), and Sarada et al. (2019) (gray dotted line). Panel C.1b reports the coverage rate, computed as the ratio between the number of patents in our dataset and the number of patents in Petrulia et al. (2016) (red line) and Sarada et al. (2019) (gray dashed line). The dataset by Sarada et al. (2019) starts in 1870; for comparability, we restrict it to “Utility” patents only. Referenced on page: A4.

Figure C.2. Example of Metropolitan Area: Chicago



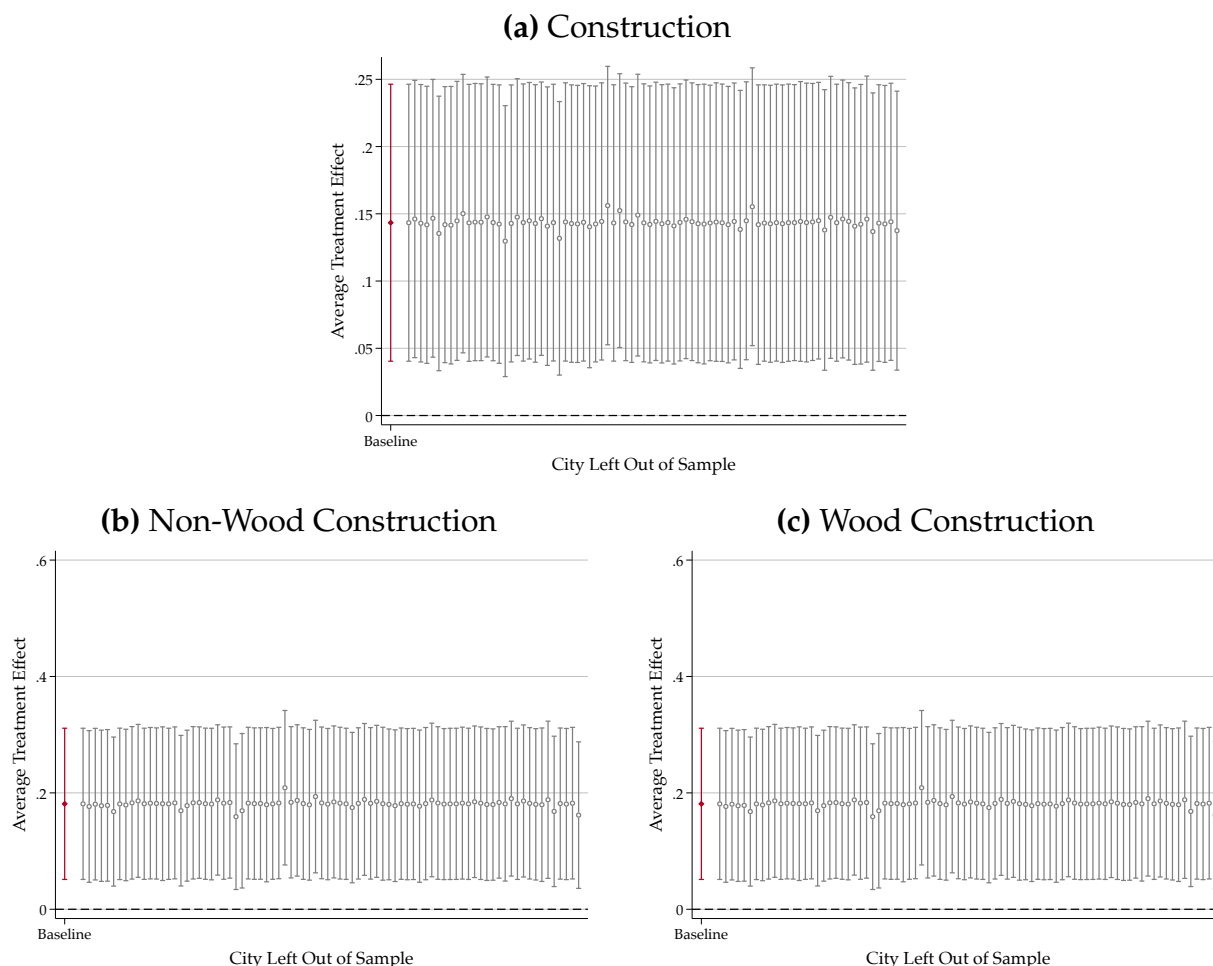
Notes. This Figure displays the construction of the Chicago Metropolitan Area according to the procedure described in Section 3.5. Each dot reports the coordinates of a single CPP location in Cook County, whose 1870 borders are displayed in the black solid line. The red star displays the location of Chicago, which is the only CPP location with more than 20,000 inhabitants in Cook County. The red dots are the minor towns—i.e., the CPP locations with less than 20,000 inhabitants—that are closer than 20 kilometers from the center of Chicago and thus are considered part of the Chicago Metropolitan Area. In this case, the Chicago Metropolitan Area includes Evanston, Lakeview, Jefferson, Cicero, Lyons, Lake, Hyde Park, and Calumet. The gray dots are towns below 20,000 inhabitants that are further than 20 kilometers from the center of Chicago and are thus excluded from its metropolitan area. Referenced on page: 10.

Figure C.3. Innovation in Construction in Chicago, Alternative Standard Errors



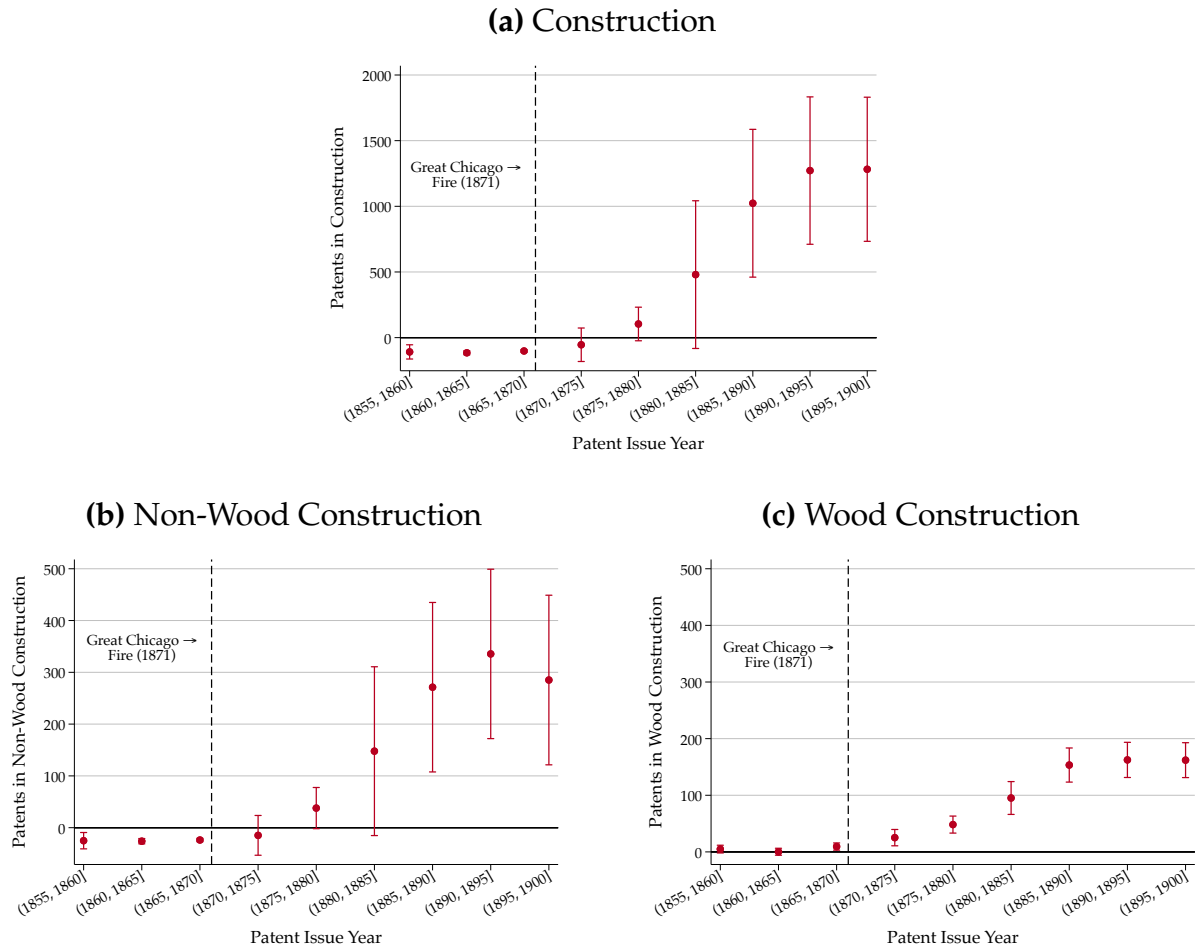
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on construction innovation. The unit of observation is a city-technology pair. Units are observed at five-year intervals between 1855 and 1900. The dependent variable is the number of patents. Each dot displays the coefficient of a treatment term defined as a triple interaction equal to one for Chicago and construction (Panel C.3a), non-wood construction (Panel C.3b), and wood construction (Panel C.3c) technology after 1871 and zero otherwise. All regressions are estimated through Poisson quasi-maximum likelihood. Regressions include city-year, city-technology, and technology-year fixed effects. Each row displays the estimated standard error obtained using different specifications: White (heteroskedasticity-adjusted), clustered at different layers, and Conley (1999) standard errors adjusting for spatial autocorrelation at various thresholds. The red dot reports the baseline estimate. Bands report 95% confidence intervals. Referenced on page: 15.

Figure C.4. Innovation in Construction in Chicago, Leave-One-Out Estimates



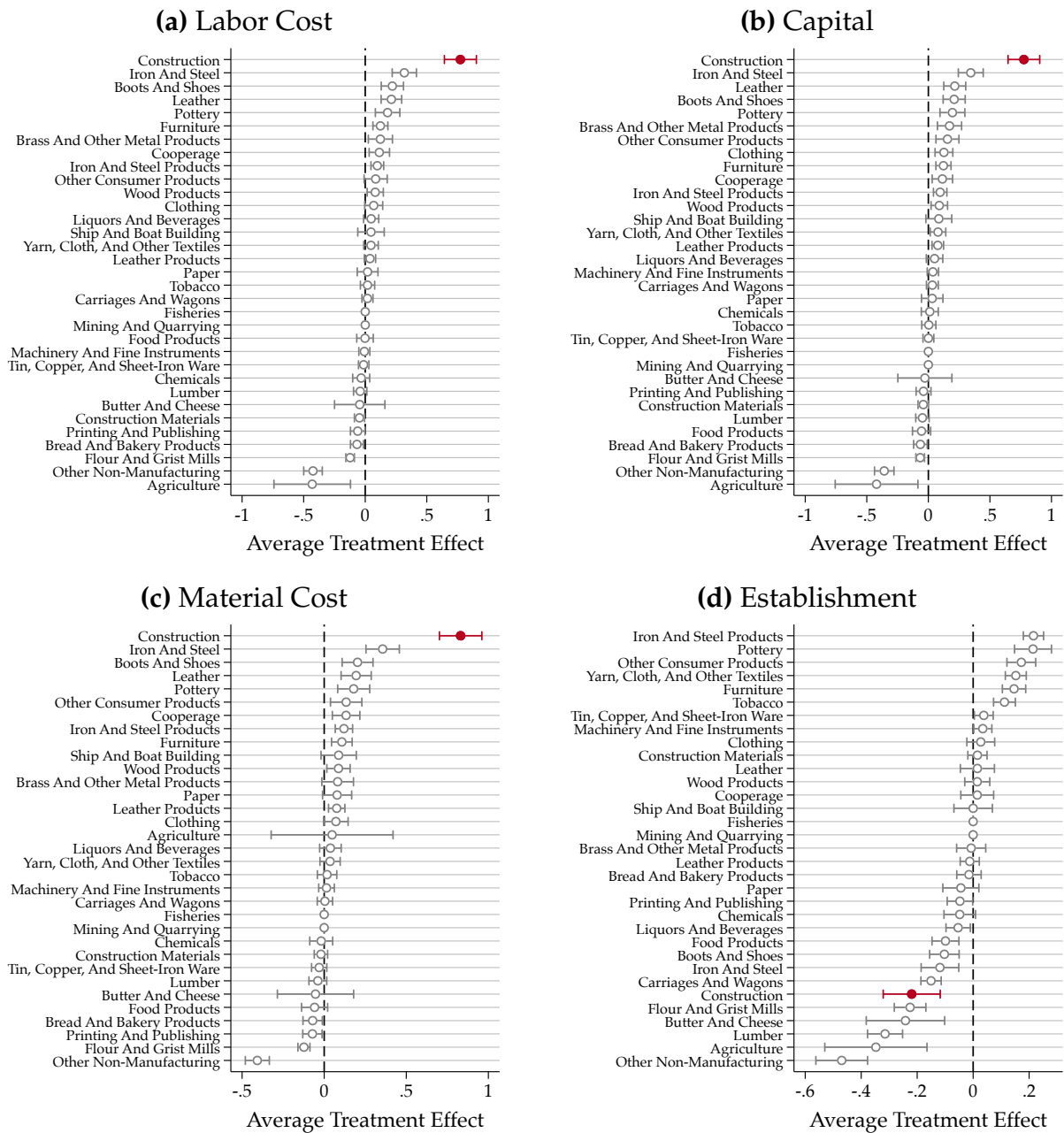
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on construction innovation. The unit of observation is a city-technology pair. Units are observed at five-year intervals between 1855 and 1900. The dependent variable is the number of patents. Each dot displays the coefficient of a treatment term defined as a triple interaction equal to one for Chicago and construction (Panel C.4a), non-wood construction (Panel C.4b), and wood construction (Panel C.4c) technology after 1871 and zero otherwise. All regressions are estimated through Poisson quasi-maximum likelihood. Regressions include city-year, city-technology, and technology-year fixed effects. Each gray dot reports the treatment effect estimated by excluding one city at a time (except Chicago) from the estimation sample; the red dot reports the baseline estimate. Standard errors are clustered at the city-technology level. Bands report 95% confidence intervals. Referenced on page: 15.

Figure C.5. Synthetic Difference-in-Differences Estimates of the Effect of the Great Chicago on Construction Innovation



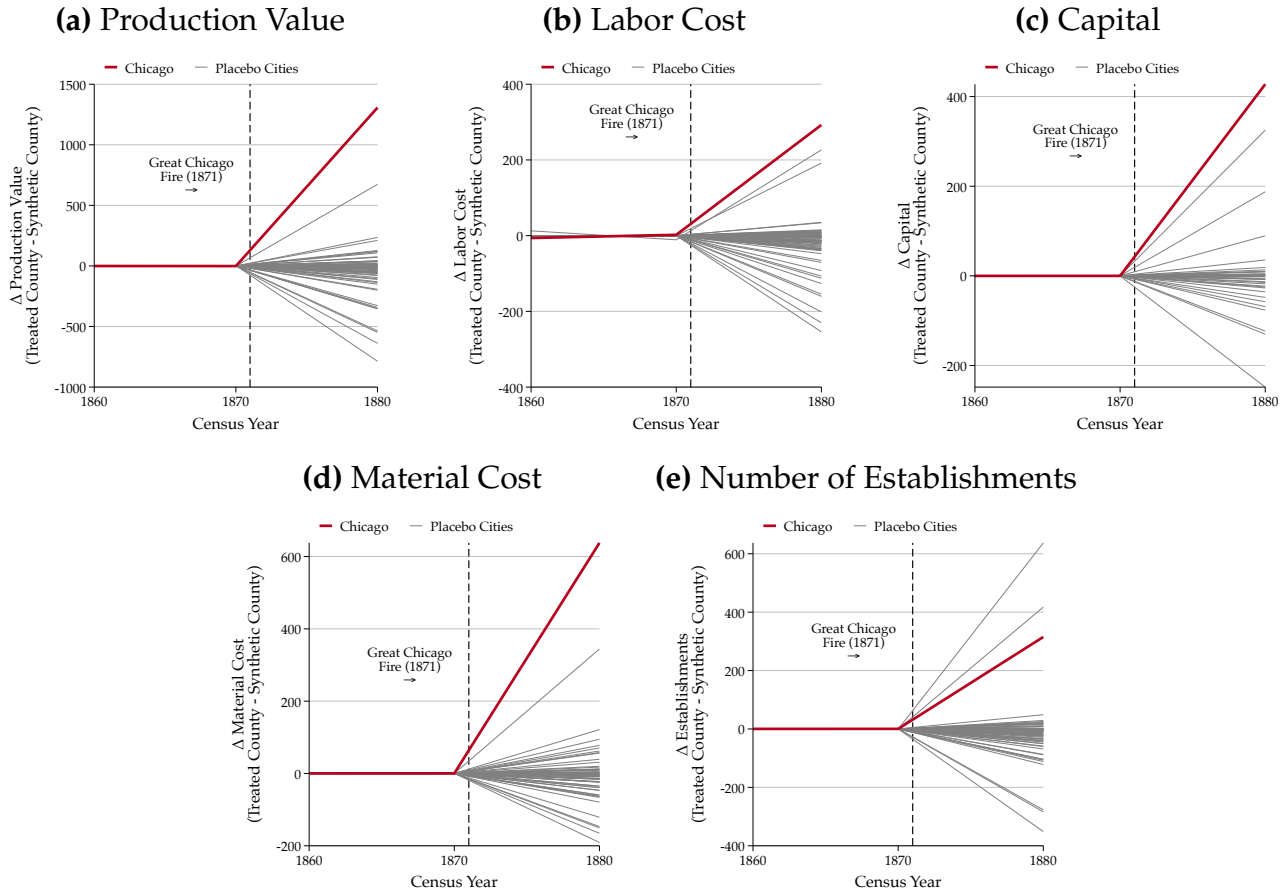
Notes. This Figure reports the effect of the Great Chicago (1871) Fire on construction (Panel C.5a), non-wood construction (Panel C.5b), and wood construction (Panel C.5c) innovation in Chicago. The dependent variable is the number of patents. The unit of observation is a metropolitan area at a five-year frequency between 1855 and 1900. The Figures report event-study coefficients obtained using the synthetic difference-in-differences methods developed by Arkhangelsky et al. (2021) and obtained following the procedure described in Clarke et al. (2023). Standard errors are estimated using bootstrap sampling with 100 replications; gray bars report 95% confidence intervals. The covariates included in the synthetic difference-in-differences estimation are the same as in the synthetic control estimation, except for the pre-treatment outcome values. The black dashed line marks the year of the Great Chicago Fire (1871). Referenced on pages: 14, 20, A9.

Figure C.6. Spillover Effects of the Great Chicago Fire on Manufacturing Activity, Additional Outcomes



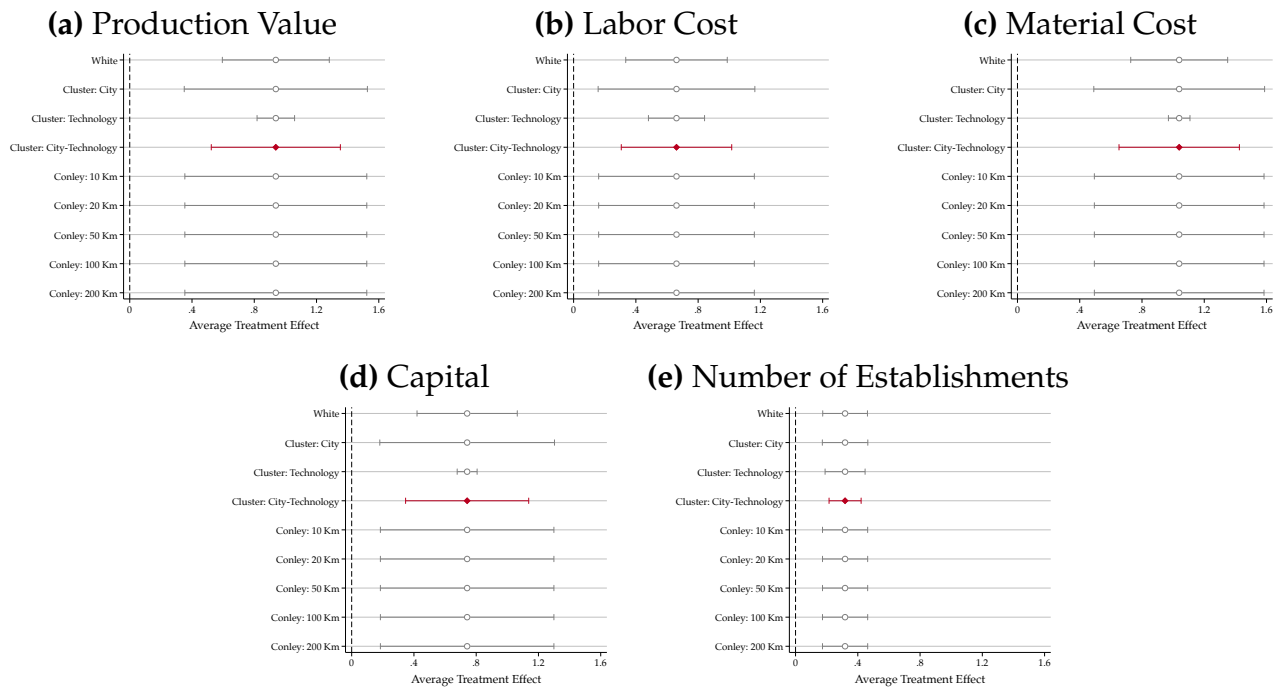
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on labor costs (Panel C.6a), capital (Panel C.6b), material costs (Panel C.6c), and the number of establishments (Panel C.6d) by industry. The unit of observation is a county-industry pair observed at a decade frequency in 1860, 1870, and 1880. All the outcome variables are expressed in log terms. For each industry, we estimate a separate regression and report the coefficient of a triple interaction equal to one for that industry in Cook County—where Chicago is located—after 1871 and zero otherwise. The model is estimated by OLS, and it includes county-year, county-industry, and industry-year fixed effects. Standard errors are clustered at the county-industry level; bands report 95% confidence intervals. We color in red the construction sector. Referenced on page: 16.

Figure C.7. Dynamic Effect of the Great Chicago Fire on Construction Manufacturing Activity: Synthetic Control Estimates



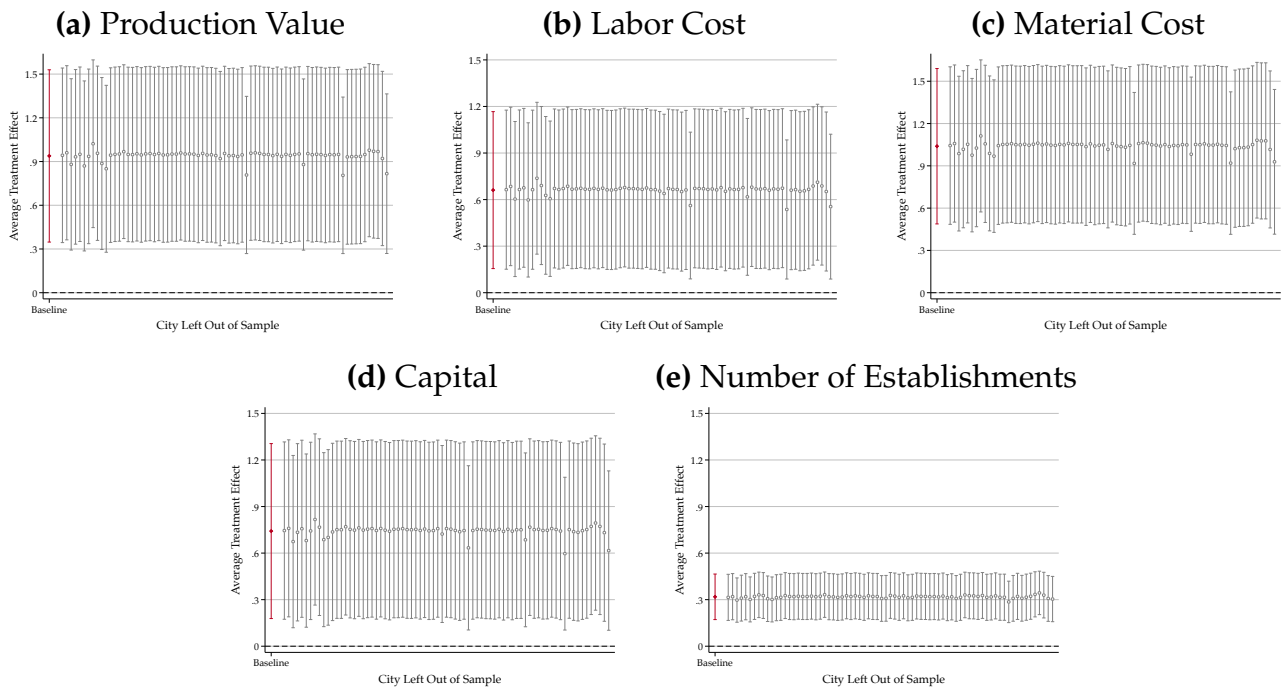
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on manufacturing activity. The unit of observation is a county observed at a decade frequency in 1860, 1870, and 1880. The dependent variable is: production value (Panel C.7a), labor cost (Panel C.7b), capital (Panel C.7c), material cost (Panel C.7d), number of establishments (Panel C.7e), all within the construction sector. In each graph, the red line reports the difference between Chicago and synthetic Chicago, and the gray line reports the difference between other cities and their synthetic counterpart. In each graph, the black dashed line reports the date of the Great Chicago Fire (1871). Referenced on page: 17.

Figure C.8. Construction Manufacturing Activity in Chicago, Alternative Standard Errors



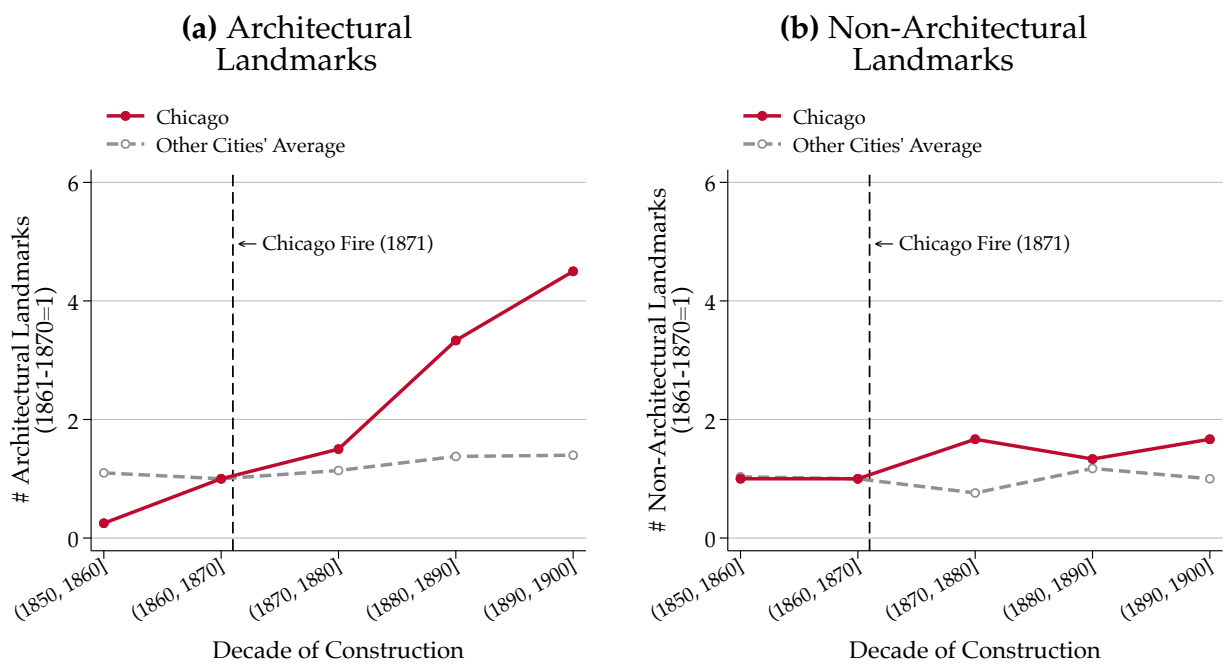
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on construction activity. The unit of observation is a county-industry pair observed at a decade frequency between 1860 and 1880. Industries are grouped into construction and other sectors. The dependent variable is: the number of production value (Panel C.8a), labor cost (Panel C.8b), material costs (Panel C.8c), capital (Panel C.8d), and the number of establishments (Panel C.8e). All dependent variables are expressed in log terms. The treatment is an interaction term equal to one for the construction sector in Cook County (where Chicago is located) in 1880, after the Fire, and zero otherwise. All regressions are estimated through OLS. Regressions include county-year, county-industry, and industry-year fixed effects. Each row displays the estimated standard error obtained using different specifications: White (heteroskedasticity-adjusted), clustered at different layers, and Conley (1999) standard errors adjusting for spatial autocorrelation at various thresholds. The red dot reports the baseline estimate. Bands report 95% confidence intervals. Referenced on page: 17.

Figure C.9. Construction Manufacturing Activity in Chicago, Leave-One-Out Estimates



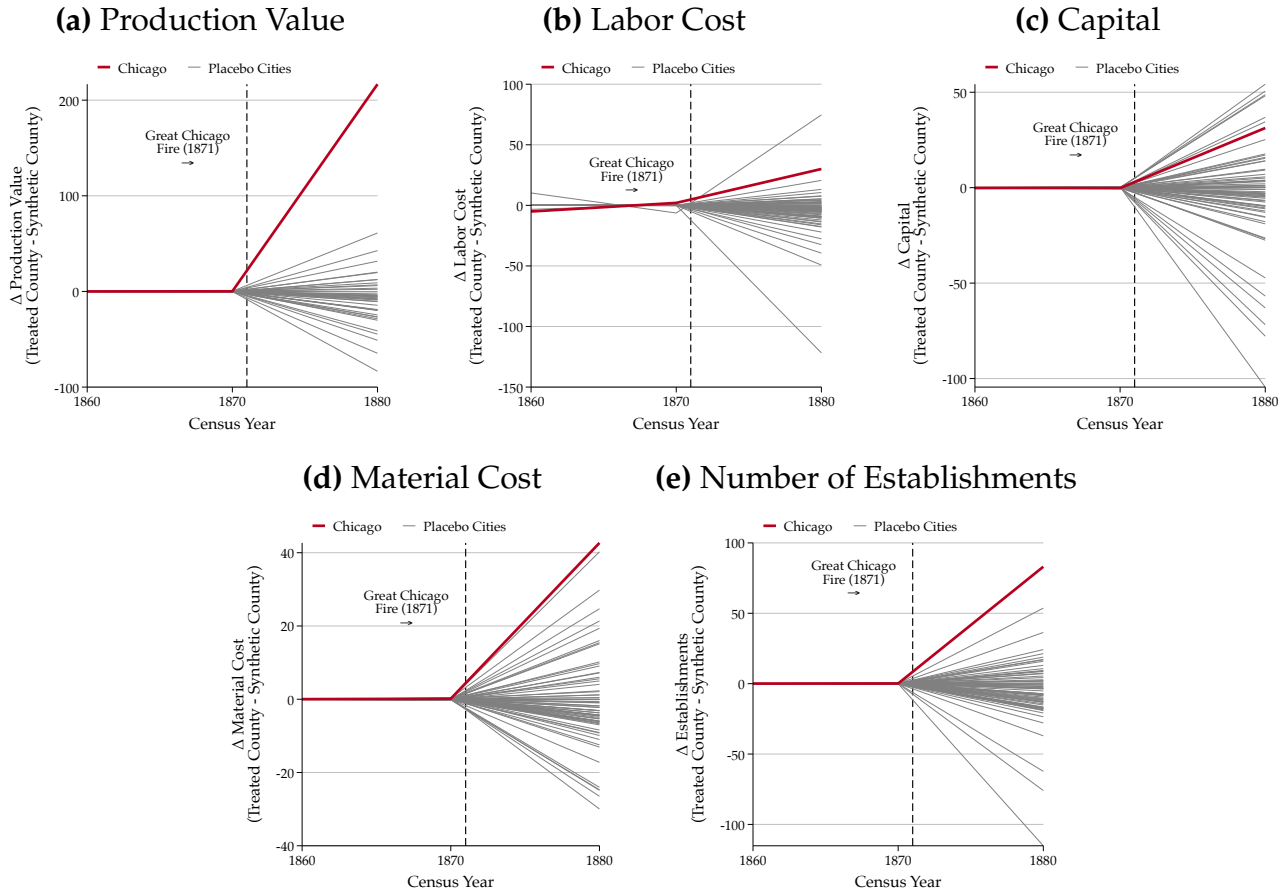
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on construction activity. The unit of observation is a county-industry pair observed at a decade frequency between 1860 and 1880. Industries are grouped into construction and other sectors. The dependent variable is: the number of production value (Panel C.9a), labor cost (Panel C.9b), material costs (Panel C.9c), capital (Panel C.9d), and the number of establishments (Panel C.9e). All dependent variables are expressed in log terms. The treatment is an interaction term equal to one for the construction sector in Cook County (where Chicago is located) in 1880, after the Fire, and zero otherwise. All regressions are estimated through OLS. Regressions include county-year, county-industry, and industry-year fixed effects. Each gray dot reports the treatment effect estimated by excluding one city at a time (except Chicago) from the estimation sample; the red dot reports the baseline estimate. Standard errors are clustered at the county-industry level. Bands report 95% confidence intervals. Referenced on page: 17.

Figure C.10. Number of Landmarks in Chicago and in Control Cities



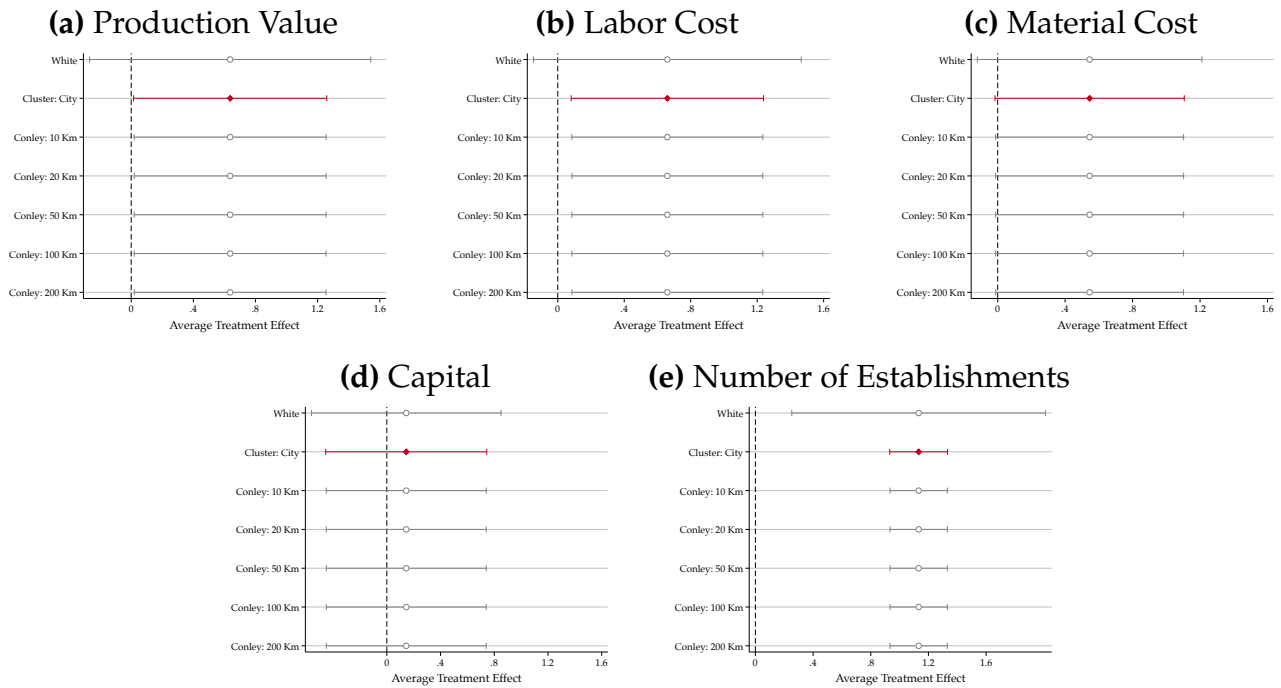
Notes. This Figure reports the number of landmark sites in Chicago and in control cities before and after the 1871 Fire. The unit of observation is a city. Cities are observed at a decade frequency between 1850 and 1900. Panel C.10a displays the number of architectural landmarks in Chicago (red) and in other cities (gray) by construction decade. Panel C.10b replicates it for non-architectural landmarks. In each panel, the number of landmarks in the 1860–1870, i.e., the last decade before the Great Chicago Fire, is normalized to one. Referenced on page: 17.

Figure C.11. Dynamic Effect of the Great Chicago Fire on Non-Wood Construction Manufacturing Activity: Synthetic Control Estimates



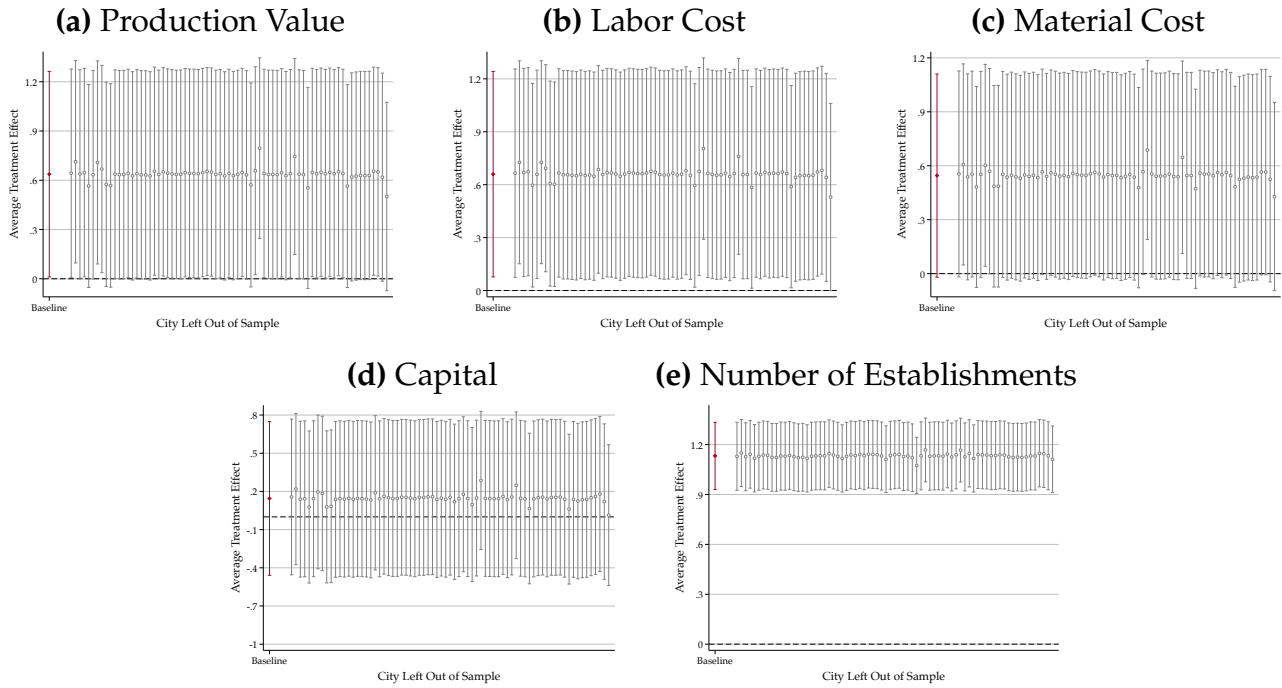
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on manufacturing activity. The unit of observation is a county observed at a decade frequency in 1860, 1870, and 1880. The dependent variable is: production value (Panel C.11a), labor cost (Panel C.11b), capital (Panel C.11c), material cost (Panel C.11d), number of establishments (Panel C.11e), all within the non-wood construction sector. In each graph, the red line reports the difference between Chicago and synthetic Chicago, and the gray line reports the difference between other cities and their synthetic counterpart. In each graph, the black dashed line reports the date of the Great Chicago Fire (1871). Referenced on page: 21.

Figure C.12. Non-Wood Construction Manufacturing Activity in Chicago, Alternative Standard Errors



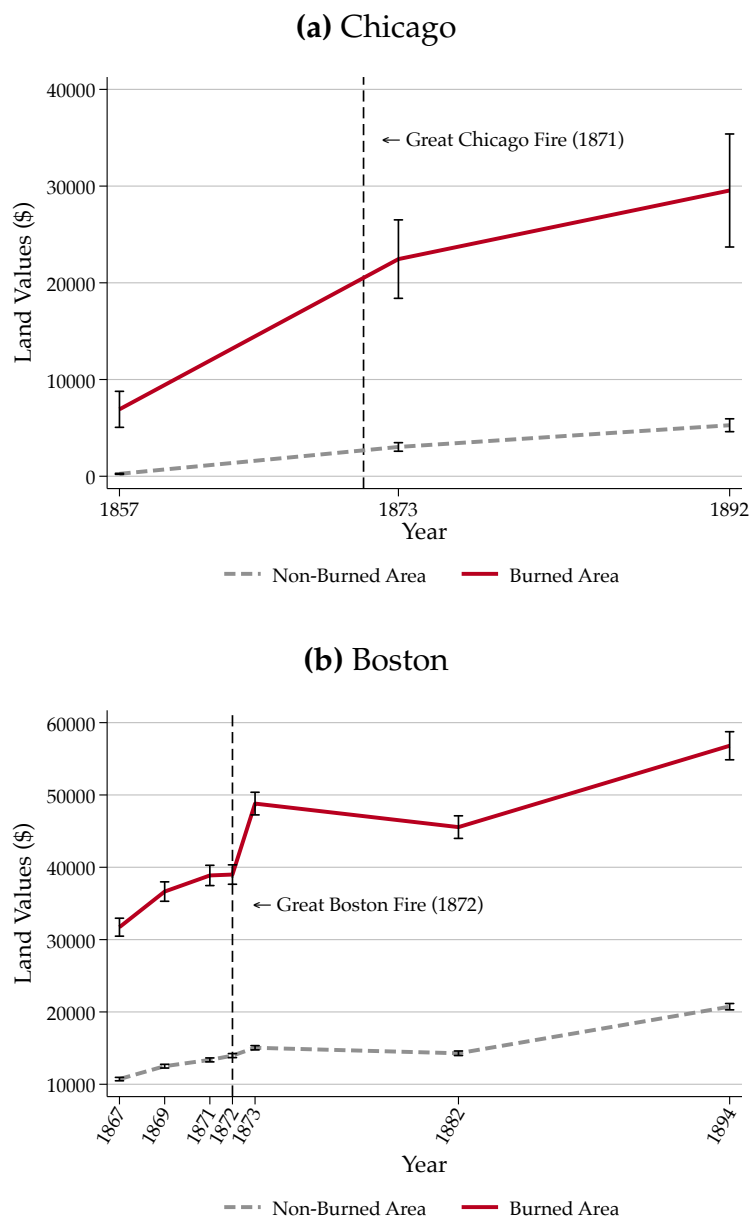
Notes. This Figure reports the effect of the Great Chicago Fire (1871) on non-wood construction activity. The unit of observation is a county observed at a decade frequency between 1860 and 1880. The dependent variable is: production value (Panel C.12a), labor costs (Panel C.12b), material costs (Panel C.12c), capital (Panel C.12d), and the number of establishments (Panel C.12e). All dependent variables refer to the non-wood industries and are expressed in log terms. The treatment is an interaction term equal to one for Cook County (where Chicago is located) in 1880, after the Fire, and zero otherwise. All regressions are estimated through OLS. Regressions include county and year fixed effects. Each row displays the estimated standard error obtained using different specifications: White (heteroskedasticity-adjusted), clustered at different layers, and Conley (1999) standard errors adjusting for spatial autocorrelation at various thresholds. The red dot reports the baseline estimate. Bands report 95% confidence intervals. Referenced on page: 21.

Figure C.13. Non-Wood Construction Manufacturing Activity in Chicago, Leave-One-Out Estimates



Notes. This Figure reports the effect of the Great Chicago Fire (1871) on non-wood construction activity. The unit of observation is a county observed at a decade frequency between 1860 and 1880. The dependent variable is: production value (Panel C.13a), labor costs (Panel C.13b), material costs (Panel C.13c), capital (Panel C.13d), and the number of establishments (Panel C.13e). All dependent variables refer to the non-wood industries and are expressed in log terms. The treatment is an interaction term equal to one for Cook County (where Chicago is located) in 1880, after the Fire, and zero otherwise. All regressions are estimated through OLS. Regressions include county and year fixed effects. Each gray dot reports the treatment effect estimated by excluding one city at a time (except Chicago) from the estimation sample; the red dot reports the baseline estimate. Standard errors are clustered at the county-industry level. Bands report 95% confidence intervals. Referenced on page: 21.

Figure C.14. Land Values in Chicago and Boston Before and After the Fires



Notes. This Figure reports the average land values per square mile in Chicago (Panel C.14a) and Boston (Panel C.14b) before and after the 1871 and 1872 fires, respectively. Data for Chicago are digitized from Hoyt (1933), while data for Boston are from Hornbeck and Keniston (2017). In each panel, the red line reports land values in areas exposed to the fires, and the gray lines report average land values for the remaining parts of the city. Bands report one-standard-deviation intervals around the mean. Referenced on page: 23.

D Additional Tables

Table D.1. List of Words Related to Construction, Wood-Related, and Non-Wood-Related Patents

Words related to...		
Construction (1)	Wood-Related Construction (2)	Non-wood Related Construction (3)
Construction	Wood	Iron
Building	Timber	Brick
Edifice	Beam	Stone
Frame	Joist	Mortar
Roof	Mortise	Concrete
Wall	Tenon	Steel
Floor	Plank	Cast
Ceiling	Bracing	Lime
Joist	Joinery	Terracotta
Pillar	Lumber	Cement
Foundation	Plywood	Glass
Footing	Veneer	
Slab		
Stair		
Staircase		
Railing		
Balustrade		
Baluster		
Fence		
Gate		
Door		
Window		
Sill		
Lintel		
Arch		
Vault		
Dome		

Notes. This Table reports the keywords that we use to implement the dictionary-based approach to identify patents in construction (column 1), wood-related construction (column 2), and non-wood-related construction (column 3). We flag a patent as construction-related if the total number of mentions of words in column (1) is five or more. A wood-related (resp. non-wood-related) patent is one that further mentions at least three times one or more words in column (2) (resp 3). Referenced on pages: [8](#), [A4](#).

Table D.2. List of Metropolitan Areas in the Sample

Metro Area (1)	City (2)	Metro Area (3)	City (4)	Metro Area (5)	City (6)
Albany	Albany	Albany	Bethlehem	Albany	Coeymans
Albany	Greenbush	Albany	Guilderland	Albany	New Baltimore
Albany	New Scotland	Albany	Niskayuna	Albany	Schodack
Allegheny	Allegheny	Allegheny	Bellevue	Allegheny	Chartiers
Allegheny	Dixmont	Allegheny	Green Tree	Allegheny	Kilbuck
Allegheny	Marshall	Allegheny	Mccandless	Allegheny	Mount Washington
Allegheny	Neville	Allegheny	Ohio	Allegheny	Peters
Allegheny	Pine	Allegheny	Richland	Allegheny	Ross
Allegheny	Scott	Allegheny	South Fayette	Allegheny	Stowe
Allegheny	Upper Saint Clair	Allegheny	Wexford		
Atlanta	Atlanta	Atlanta	Clayton	Atlanta	Decatur
Atlanta	Dekalb	Atlanta	Panthersville		
Augusta	Augusta	Augusta	Hamburg	Augusta	Richmond Factory Pond
Baltimore	Baltimore	Baltimore	Baltimore Zoo	Baltimore	Brooklandville
Baltimore	Brooklyn	Baltimore	Catonsville	Baltimore	Cockeysville
Baltimore	Lutherville	Baltimore	Saint Denis	Baltimore	Texas
Baltimore	Warren				
Boston	Boston	Boston	Braintree	Boston	Milton
Boston	Quincy	Boston	Randolph	Boston	Stoughton
Boston	Weymouth				
Brooklyn	Brooklyn	Brooklyn	College Point	Brooklyn	Columbusville
Brooklyn	Flatbush	Brooklyn	Flatlands	Brooklyn	Flushing
Brooklyn	Gravesend	Brooklyn	Jamaica	Brooklyn	New Lots
Brooklyn	New Utrecht	Brooklyn	Westchester	Brooklyn	Whitestone
Buffalo	Amherst	Buffalo	Buffalo	Buffalo	Cheektowaga
Buffalo	Eden	Buffalo	Grand Island	Buffalo	Hamburg
Buffalo	Tonawanda	Buffalo	West Seneca	Buffalo	Wheatfield
Buffalo	Williamsville				
Camden	Blackwood	Camden	Camden	Camden	Cinnaminson
Camden	Deptford	Camden	Gloucester	Camden	Gloucester City
Camden	Haddon	Camden	Haddonfield	Camden	Harrison
Camden	Mantua	Camden	Merchantville	Camden	Washington
Camden	Woodbury				
Charleston	Charleston	Charleston	James Island	Charleston	Johns Island
Charleston	Saint Andrews				
Charlestown	Charlestown	Charlestown	Chelsea	Charlestown	East Boston
Charlestown	Everett	Charlestown	Malden	Charlestown	Melrose
Charlestown	Wakefield				
Chatham	Chatham	Chatham	Whitmell		
Chicago	Calumet	Chicago	Chicago	Chicago	Cicero
Chicago	Evanston	Chicago	Hyde Park	Chicago	Jefferson

Chicago	Lake	Chicago	Lakeview	Chicago	Lyons
Cincinnati	Bromley	Cincinnati	Cincinnati	Cincinnati	Colerain
Cincinnati	Cumminsville	Cincinnati	Delhi	Cincinnati	Glendale
Cincinnati	Green	Cincinnati	Ludlow	Cincinnati	Millcreek
Cincinnati	Springdale	Cincinnati	Springfield	Cincinnati	Taylorsport
Cincinnati	West Covington				
Cleveland	Bedford	Cleveland	Brecksville	Cleveland	Brooklyn
Cleveland	Cleveland	Cleveland	East Cleveland	Cleveland	Independence
Cleveland	Newburg	Cleveland	Parma	Cleveland	Rockport
Cleveland	Royalton				
Columbus	Blendon	Columbus	Clinton	Columbus	Columbus
Columbus	Franklin	Columbus	Groveport	Columbus	Hamilton
Columbus	Harrison	Columbus	Jackson	Columbus	Madison
Columbus	Mifflin	Columbus	Norwich	Columbus	Orange
Columbus	Perry	Columbus	Scioto	Columbus	Sharon
Columbus	Truro	Columbus	Westerville		
Covington	Alexandria	Covington	Anderson	Covington	Bank Lick
Covington	Cold Spring	Covington	Columbia	Covington	Covington
Covington	Dayton	Covington	Florence	Covington	Independence
Covington	Johns Hill	Covington	Kenton	Covington	Newport
Covington	Pendleton	Covington	Scott	Covington	Visalia
Davenport	Andalusia	Davenport	Blackhawk	Davenport	Bowling
Davenport	Buffalo	Davenport	Butler	Davenport	Coal Valley
Davenport	Davenport	Davenport	Hampton	Davenport	Lincoln
Davenport	Moline	Davenport	Pleasant Valley	Davenport	Preemption
Davenport	Rock Island	Davenport	Rockingham	Davenport	Sheridan
Davenport	Winfield				
Dayton	Bath	Dayton	Beavercreek	Dayton	Bellbrook
Dayton	Bethel	Dayton	Brandt	Dayton	Butler
Dayton	Clear Creek	Dayton	Dayton	Dayton	Harrison
Dayton	Jefferson	Dayton	Mad River	Dayton	Madison
Dayton	Miami	Dayton	Miamisburg	Dayton	Monroe
Dayton	Randolph	Dayton	Sugarcreek	Dayton	Vandalia
Dayton	Washington	Dayton	Wayne	Dayton	West Charleston
Detroit	Dearborn	Detroit	Detroit	Detroit	Ecorse
Detroit	Fort Wayne	Detroit	Grosse Pointe	Detroit	Hamtramck
Detroit	Roseville	Detroit	Royal Oak	Detroit	Springwells
Detroit	Warren	Detroit	Wyandotte		
Edgefield	Edgefield	Edgefield	Fruit Hill	Edgefield	Johnston
Edgefield	Meeting Street				
Elizabeth	Clark	Elizabeth	Elizabeth	Elizabeth	Keyport
Elizabeth	Linden	Elizabeth	Livingston	Elizabeth	Millburn
Elizabeth	New Dorp	Elizabeth	New Springville	Elizabeth	Perth Amboy
Elizabeth	Port Richmond	Elizabeth	Rahway	Elizabeth	Richmond
Elizabeth	South Amboy	Elizabeth	South Orange	Elizabeth	Springfield
Elizabeth	Summit	Elizabeth	Tottenville	Elizabeth	Westfield
Elizabeth	Woodbridge				
Eufaula	Eufaula	Eufaula	Georgetown		
Evansville	Armstrong	Evansville	Campbell	Evansville	Center

Evansville	Evansville	Evansville	German	Evansville	Henderson
Evansville	Knight	Evansville	Ohio	Evansville	Perry
Evansville	Pigeon	Evansville	Scott	Evansville	Spottsville
Evansville	Union				
Fall River	Berkley	Fall River	Bristol	Fall River	Dighton
Fall River	Fall River	Fall River	Little Compton	Fall River	Middletown
Fall River	Portsmouth	Fall River	Rehoboth	Fall River	Somerset
Fall River	Swansea	Fall River	Taunton	Fall River	Tiverton
Fall River	Warren	Fall River	Westport		
Harrisburg	Dauphin	Harrisburg	Duncannon	Harrisburg	East Pennsboro
Harrisburg	Fairview	Harrisburg	Goldsboro	Harrisburg	Halifax
Harrisburg	Hampden	Harrisburg	Harrisburg	Harrisburg	Highspire
Harrisburg	Hummelstown	Harrisburg	Lewisberry	Harrisburg	Lower Allen
Harrisburg	Lower Paxton	Harrisburg	Lower Swatara	Harrisburg	Marysville
Harrisburg	Mechanicsburg	Harrisburg	Middle Paxton	Harrisburg	Middletown
Harrisburg	Monaghan	Harrisburg	New Buffalo	Harrisburg	New
					Cumberland
Harrisburg	Newberry	Harrisburg	Penn	Harrisburg	Reed
Harrisburg	Rockville	Harrisburg	Rye	Harrisburg	Silver Spring
Harrisburg	South Hanover	Harrisburg	Susquehanna	Harrisburg	Upper Allen
Harrisburg	Warrington	Harrisburg	Watts	Harrisburg	West Hanover
Hartford	Avon	Hartford	Berlin	Hartford	Bloomfield
Hartford	Cromwell	Hartford	East Granby	Hartford	East Hartford
Hartford	East Windsor	Hartford	Farmington	Hartford	Glastonbury
	Hill				
Hartford	Granby	Hartford	Hartford	Hartford	Manchester
Hartford	Middletown	Hartford	New Britain	Hartford	Portland
Hartford	Rocky Hill	Hartford	Simsbury	Hartford	South Windsor
Hartford	West Hartford	Hartford	Wethersfield	Hartford	Windsor
Hoboken	Hackensack	Hoboken	Hoboken	Hoboken	North Bergen
Hoboken	Weehawken				
Indianapolis	Allisonville	Indianapolis	Carmel	Indianapolis	Center
Indianapolis	Clay	Indianapolis	Decatur	Indianapolis	Eagle
Indianapolis	Franklin	Indianapolis	Indianapolis	Indianapolis	Lawrence
Indianapolis	Millersville	Indianapolis	Perry	Indianapolis	Pike
Indianapolis	Pleasant	Indianapolis	Warren	Indianapolis	Washington
Indianapolis	Wayne	Indianapolis	White River		
Jersey City	Bayonne	Jersey City	Greenville	Jersey City	Jersey City
Jersey City	New Brighton	Jersey City	Rutherford Park	Jersey City	Tompkinsville
Kansas City	Gallatin	Kansas City	Independence	Kansas City	Kansas City
Kansas City	Pettis	Kansas City	Quindaro	Kansas City	Shawnee
Kansas City	Westport				
Knoxville	Beaver Ridge	Knoxville	Knoxville	Knoxville	Louisville
Knoxville	Maryville	Knoxville	Rockford		
Lancaster	Clay	Lancaster	Conestoga	Lancaster	East Hempfield
Lancaster	East Lampeter	Lancaster	Eden	Lancaster	Elizabeth
Lancaster	Ephrata	Lancaster	Lancaster	Lancaster	Leacock
Lancaster	Manheim	Lancaster	Manor	Lancaster	Martic

Lancaster	Millersville	Lancaster	New Providence	Lancaster	Paradise
Lancaster	Penn	Lancaster	Pequea	Lancaster	Rapho
Lancaster	Strasburg	Lancaster	Upper Leacock	Lancaster	Warwick
Lancaster	West Earl	Lancaster	West Hempfield	Lancaster	West Lampeter
Lawrence	Andover	Lawrence	Atkinson	Lawrence	Bradford
Lawrence	Danville	Lawrence	Georgetown	Lawrence	Groveland
Lawrence	Hampstead	Lawrence	Haverhill	Lawrence	Lawrence
Lawrence	Methuen	Lawrence	Newton	Lawrence	North Andover
Lawrence	North Reading	Lawrence	Plaistow	Lawrence	Salem
Lawrence	Sandown	Lawrence	Wilmington		
Louisville	Carr	Louisville	Charlestown	Louisville	Franklin
Louisville	Harrods Creek	Louisville	Jeffersonville	Louisville	Lafayette
Louisville	Louisville	Louisville	New Albany	Louisville	Newburg
Louisville	Saint Matthews	Louisville	Shively	Louisville	Silver Creek
Louisville	Springdale	Louisville	Union	Louisville	Utica
Louisville	nan				
Lowell	Acton	Lowell	Bedford	Lowell	Billerica
Lowell	Carlisle	Lowell	Chelmsford	Lowell	Concord
Lowell	Dracut	Lowell	Dunstable	Lowell	Hudson
Lowell	Lincoln	Lowell	Lowell	Lowell	Pelham
Lowell	Tewksbury	Lowell	Tyngsborough	Lowell	Westford
Lowell	Windham				
Lynchburg	Amherst	Lynchburg	Brookville	Lynchburg	Coolwell
Lynchburg	Elon	Lynchburg	Forest	Lynchburg	Lynchburg
Lynchburg	New London				
Lynn	Hingham	Lynn	Hull	Lynn	Lynn
Lynn	Lynnfield	Lynn	Middleton	Lynn	Nahant
Lynn	Saugus	Lynn	Swampscott	Lynn	Winthrop
Macon	Clinton	Macon	Macon		
Manchester	Allenstown	Manchester	Auburn	Manchester	Bedford
Manchester	Bow	Manchester	Concord	Manchester	Derry
Manchester	Dunbarton	Manchester	Goffstown	Manchester	Hooksett
Manchester	Litchfield	Manchester	Londonderry	Manchester	Manchester
Manchester	Merrimack	Manchester	Nashua	Manchester	Pembroke
Marion	Marion	Marion	Perry		
Memphis	Cuba	Memphis	Desoto	Memphis	Hopefield
Memphis	Horn Lake	Memphis	Memphis		
Milwaukee	Caledonia	Milwaukee	Franklin	Milwaukee	Granville
Milwaukee	Greenfield	Milwaukee	Lake	Milwaukee	Mequon
Milwaukee	Milwaukee	Milwaukee	Oak Creek	Milwaukee	Wauwatosa
Montgomery	Autauga	Montgomery	Elmore	Montgomery	Montgomery
Montgomery	Prattville	Montgomery	Wetumpka		
Nashville	Brentwood	Nashville	Goodlettsville	Nashville	Madison
Nashville	Nashville				
New Bedford	Acushnet	New Bedford	Dartmouth	New Bedford	Fairhaven
New Bedford	Freetown	New Bedford	Gosnold	New Bedford	Lakeville
New Bedford	Marion	New Bedford	Mattapoisett	New Bedford	New Bedford
New Bedford	Rochester				
New Haven	Ansonia	New Haven	Bethany	New Haven	Branford

New Haven	Cheshire	New Haven	Derby	New Haven	East Haven
New Haven	Hamden	New Haven	Milford	New Haven	New Haven
New Haven	North Branford	New Haven	North Haven	New Haven	Orange
New Haven	Prospect	New Haven	Seymour	New Haven	Wallingford
New Haven	West Haven	New Haven	Woodbridge		
New Orleans	Carrollton	New Orleans	Kenner	New Orleans	Metairie
New Orleans	New Orleans	New Orleans	Shrewsbury		
New York	Astoria	New York	Belmont	New York	Fordham
New York	Long Island City	New York	Morrisania	New York	New York
New York	Tremont	New York	West Farms		
Newark	Belleville	Newark	Bloomfield	Newark	East Orange
Newark	Harrison	Newark	Kearny	Newark	Montclair
Newark	Newark	Newark	Orange	Newark	West Orange
Newburgh	Cold Spring	Newburgh	Cornwall	Newburgh	Fishkill
Newburgh	Fishkill Landing	Newburgh	Glenham	Newburgh	New Windsor
Newburgh	Newburgh	Newburgh	Philipstown	Newburgh	Plattekill
Newburgh	West Point				
Norfolk	Deep Creek	Norfolk	Fort Monroe	Norfolk	Great Bridge
Norfolk	Hampton	Norfolk	Kempsville	Norfolk	Norfolk
Norfolk	Old Point	Norfolk	Portsmouth	Norfolk	Tanner Creek
	Comfort Marina				
Norfolk	Western Branch	Norfolk	Wythe		
	Park				
North	Bellingham	North	Blackstone	North	Cumberland
Providence		Providence		Providence	
North	Johnston	North	North	North	Scituate
Providence		Providence	Providence	Providence	
North	Smithfield	North	Woonsocket		
Providence		Providence			
Old Cambridge	Arlington	Old Cambridge	Belmont	Old Cambridge	Brighton
Old Cambridge	Brookline	Old Cambridge	Burlington	Old Cambridge	Canton
Old Cambridge	Dedham	Old Cambridge	Hyde Park	Old Cambridge	Lexington
Old Cambridge	Medford	Old Cambridge	Needham	Old Cambridge	Newton
Old Cambridge	Old Cambridge	Old Cambridge	Reading	Old Cambridge	Somerville
Old Cambridge	Stoneham	Old Cambridge	Waltham	Old Cambridge	Watertown
Old Cambridge	West Roxbury	Old Cambridge	Winchester	Old Cambridge	Woburn
Oswego	Fulton	Oswego	Granby	Oswego	Hannibal
Oswego	Ira	Oswego	Martville	Oswego	Oswego
Oswego	Scriba	Oswego	Sterling	Oswego	Sterling Valley
Oswego	Volney				
Paterson	Caldwell	Paterson	Clinton	Paterson	Hohokus
Paterson	Little Falls	Paterson	Lodi	Paterson	Passaic
Paterson	Paterson	Paterson	Pequannock	Paterson	Pompton
Paterson	Ramapo	Paterson	Ramsey	Paterson	Saddle River
Paterson	Washington	Paterson	Wayne		
Peoria	Akron	Peoria	Cincinnati	Peoria	Dillon
Peoria	Elm Grove	Peoria	Fondulac	Peoria	Groveland
Peoria	Hallock	Peoria	Hollis	Peoria	Kickapoo

Peoria	Limestone	Peoria	Medina	Peoria	Morton
Peoria	Pekin	Peoria	Peoria	Peoria	Radnor
Peoria	Richwoods	Peoria	Spring Bay	Peoria	Tremont
Peoria	Worth				
Philadelphia	Abington	Philadelphia	Cheltenham	Philadelphia	Conshohocken
Philadelphia	Darby	Philadelphia	Greenwich	Philadelphia	Haverford
Philadelphia	Horsham	Philadelphia	Lower Merion	Philadelphia	Oreland
Philadelphia	Philadelphia	Philadelphia	Ridley	Philadelphia	Springfield
Philadelphia	Springfield	Philadelphia	Tinicum	Philadelphia	Upper Darby
Philadelphia	Upper Dublin	Philadelphia	Whitemarsh		
Pittsburgh	Allentown	Pittsburgh	Baldwin	Pittsburgh	Braddock
Pittsburgh	East Pittsburgh	Pittsburgh	Elizabeth	Pittsburgh	Etna
Pittsburgh	Hampton	Pittsburgh	Indiana	Pittsburgh	Lincoln
Pittsburgh	Mckeesport	Pittsburgh	Mifflin	Pittsburgh	Millvale
Pittsburgh	Pittsburgh	Pittsburgh	Reserve	Pittsburgh	Shaler
Pittsburgh	Sharpsburg	Pittsburgh	Snowden	Pittsburgh	Surgeon Hall
Pittsburgh	Union	Pittsburgh	West Elizabeth	Pittsburgh	Wilkins
Portland	Cape Elizabeth	Portland	Cumberland	Portland	Falmouth
Portland	Gray	Portland	North Yarmouth	Portland	Portland
Portland	Scarborough	Portland	Westbrook	Portland	Yarmouth
Poughkeepsie	Clinton	Poughkeepsie	East Fishkill	Poughkeepsie	Esopus
Poughkeepsie	Fishkill Plains	Poughkeepsie	Hyde Park	Poughkeepsie	La Grange
Poughkeepsie	Lloyd	Poughkeepsie	Marlborough	Poughkeepsie	New Paltz
Poughkeepsie	Pleasant Valley	Poughkeepsie	Port Ewen	Poughkeepsie	Poughkeepsie
Poughkeepsie	Rhinebeck	Poughkeepsie	Rondout	Poughkeepsie	Sleightsburg
Poughkeepsie	Wappingers				
	Falls				
Providence	Attleboro	Providence	Barrington	Providence	Cranston
Providence	East Greenwich	Providence	East Providence	Providence	Pawtucket
Providence	Providence	Providence	Seekonk	Providence	Warwick
Quincy	Burton	Quincy	Ellington	Quincy	Fabius
Quincy	Fall Creek	Quincy	La Grange	Quincy	Liberty
Quincy	Melrose	Quincy	Mendon	Quincy	Miller
Quincy	Palmyra	Quincy	Quincy	Quincy	South River
Quincy	Ursa				
Reading	Adamstown	Reading	Alsace	Reading	Bern
Reading	Brecknock	Reading	Brecknock	Reading	Caernarvon
Reading	Caernarvon	Reading	Centre	Reading	Cumru
Reading	Exeter	Reading	Hamburg	Reading	Maiden Creek
Reading	Muhlenberg	Reading	Oley	Reading	Ontelaunee
Reading	Penn	Reading	Perry	Reading	Reading
Reading	Richmond	Reading	Robeson	Reading	Ruscombmanor
Reading	South	Reading	Spring	Reading	Union
	Heidelberg				
Richmond	Bermuda	Richmond	Brookland	Richmond	Chester
Richmond	Chesterfield	Richmond	Clover Hill	Richmond	Dale
Richmond	Fairfield	Richmond	Manchester	Richmond	Richmond
Richmond	Tuckahoe	Richmond	Varina		

Rochester	Brighton	Rochester	Chili	Rochester	Gates
Rochester	Greece	Rochester	Henrietta	Rochester	Honeoye Falls
Rochester	Irondequoit	Rochester	Mendon	Rochester	Penfield
Rochester	Pittsford	Rochester	Rochester	Rochester	Rush
Rochester	Scottsville				
Sacramento	Brighton	Sacramento	Franklin	Sacramento	Fremont
Sacramento	Sacramento				
Saint Louis	Brooklyn	Saint Louis	Carondelet	Saint Louis	Caseyville
Saint Louis	Columbia	Saint Louis	East Saint Louis	Saint Louis	French
Saint Louis	Gartside	Saint Louis	Madison	Saint Louis	Millstadt
Saint Louis	Saint Ferdinand	Saint Louis	Saint Louis	Saint Louis	Venice
Saint Paul	Centerville	Saint Paul	Cottage Grove	Saint Paul	Eagan
Saint Paul	Inver Grove	Saint Paul	Mendota	Saint Paul	Minneapolis
Saint Paul	Mounds View	Saint Paul	Newport	Saint Paul	Oakdale
Saint Paul	Rosemount	Saint Paul	Roseville	Saint Paul	Saint Anthony
Saint Paul	Saint Paul	Saint Paul	West Saint Paul	Saint Paul	White Bear
Saint Paul	Woodbury				
Salem	Beverly	Salem	Boxford	Salem	Danvers
Salem	Essex	Salem	Hamilton	Salem	Ipswich
Salem	Manchester	Salem	Marblehead	Salem	Peabody
Salem	Rowley	Salem	Salem	Salem	Topsfield
Salem	Wenham				
San Francisco	Alameda	San Francisco	Oakland	San Francisco	San Bruno
San Francisco	San Francisco	San Francisco	San Pablo	San Francisco	Sausalito
Savannah	Hardeeville	Savannah	Savannah	Savannah	Thunderbolt
Savannah	White Bluff				
Scranton	Archbald	Scranton	Benton	Scranton	Blakely
Scranton	Dunmore	Scranton	Exeter	Scranton	Greenfield
Scranton	Hyde Park	Scranton	Jefferson	Scranton	Jenkins
Scranton	North Abington	Scranton	Old Forge	Scranton	Olyphant
Scranton	Pittston	Scranton	Ransom	Scranton	Scott
Scranton	Scranton	Scranton	South Abington	Scranton	Spring Brook
Scranton	Waverly	Scranton	West Pittston		
Shreveport	Bossier City	Shreveport	Shreveport		
Springfield	Agawam	Springfield	Chicopee	Springfield	Easthampton
Springfield	Enfield	Springfield	Granby	Springfield	Hadley
Springfield	Holyoke	Springfield	Longmeadow	Springfield	Ludlow
Springfield	Northampton	Springfield	Somers	Springfield	South Hadley
Springfield	Southampton	Springfield	Springfield	Springfield	Suffield
Springfield	West Springfield	Springfield	Westfield	Springfield	Wilbraham
Springfield	Windsor Locks				
Syracuse	Amboy	Syracuse	Belgium	Syracuse	Belleisle
Syracuse	Brewerton	Syracuse	Camillus	Syracuse	Cardiff
Syracuse	Caughdenoy	Syracuse	Central Square	Syracuse	Cicero
Syracuse	Clay	Syracuse	Dewitt	Syracuse	Euclid
Syracuse	Fayetteville	Syracuse	Geddes	Syracuse	Jamesville
Syracuse	Lafayette	Syracuse	Liverpool	Syracuse	Manlius
Syracuse	Navarino	Syracuse	Onondaga	Syracuse	Otisco
Syracuse	Pompey	Syracuse	Salina	Syracuse	South Onondaga

Syracuse	Syracuse	Syracuse	Three Rivers	Syracuse	Van Buren
Toledo	Bedford	Toledo	Erie	Toledo	Lake
Toledo	Maumee	Toledo	Oregon	Toledo	Perrysburg
Toledo	Sylvania	Toledo	Toledo	Toledo	Troy
Toledo	Washington	Toledo	Webster	Toledo	Whiteford
Trenton	Bordentown	Trenton	Bristol	Trenton	Burlington
Trenton	Chesterfield	Trenton	Ewing	Trenton	Falls
Trenton	Hamilton	Trenton	Hamilton Square	Trenton	Hopewell
Trenton	Lawrence	Trenton	Lower	Trenton	Mansfield
			Makefield		
Trenton	Middletown	Trenton	Morrisville	Trenton	Mount Holly
Trenton	Princeton	Trenton	Springfield	Trenton	Trenton
Trenton	West Windsor	Trenton	Westampton		
Troy	Brunswick	Troy	Clifton Park	Troy	Cohoes
Troy	Green Island	Troy	Halfmoon	Troy	Lansingburgh
Troy	Mechanicville	Troy	Nassau	Troy	North
					Greenbush
Troy	Poestenkill	Troy	Sand Lake	Troy	Schaghticoke
Troy	Stillwater	Troy	Troy	Troy	Waterford
Troy	Watervliet	Troy	West Sand Lake		
Utica	Bridgewater	Utica	Cassville	Utica	Clark Mills
Utica	Clayville	Utica	Clinton	Utica	Deerfield
Utica	Floyd	Utica	Frankfort	Utica	Gravesville
Utica	Holland Patent	Utica	Kirkland	Utica	Litchfield
Utica	Marcy	Utica	Marshall	Utica	New Hartford
Utica	New York Mills	Utica	Oriskany	Utica	Paris
Utica	Prospect	Utica	Remsen	Utica	Sauquoit
Utica	Schuyler	Utica	South Trenton	Utica	Steuben

Notes. This Table reports the list of metropolitan areas (columns 1, 3, and 5) and all the cities below 20,000 inhabitants that are part of them. There is a total of 84 metropolitan areas comprising 1,048 smaller towns. Referenced on page: [10](#).

Table D.3. The Great Chicago and Boston Fires and Construction Manufacturing, Entire United States

	Number of Establishments	Production Value	Labor Costs	Material Costs	Capital
	(1)	(2)	(3)	(4)	(5)
Panel A: Cook County					
Cook County $\times$ Construction $\times$ Post-1871	0.559*** (0.041)	2.231*** (0.188)	1.649*** (0.160)	2.192*** (0.173)	1.979*** (0.177)
Mean Dep. Var.	2.180	8.967	7.622	8.410	8.453
Panel B: Suffolk County					
Suffolk County $\times$ Construction $\times$ Post-1872	0.372*** (0.034)	2.036*** (0.158)	1.524*** (0.134)	1.745*** (0.145)	1.679*** (0.148)
Mean Dep. Var.	2.422	9.369	8.091	8.754	8.846
County-Year FE	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes
County-Industry FE	Yes	Yes	Yes	Yes	Yes
# Counties	2,086	2,086	2,086	2,086	2,086
# Observations	9,300	9,300	9,300	9,300	9,300

Notes. This Table reports the effect of the Great Chicago (1871) and Boston (1872) Fires on construction activity. The unit of observation is a county-industry pair observed at a decade frequency between 1860 and 1880. Industries are grouped into construction and other sectors. The sample includes all counties in the United States. The dependent variable is: the number of establishments (1), production value (2), labor costs (3), material costs (4), and capital (5). All dependent variables are expressed in log terms. The treatment is an interaction term equal to one for the construction sector in Cook County (where Chicago is located, in Panel A) or Suffolk County (where Boston is located, in Panel B) in 1880, after their respective fires, and zero otherwise. All regressions are estimated through OLS. Regressions include county-year, county-industry, and industry-year fixed effects. Standard errors are clustered at the county-industry level. Referenced on pages: [16](#), [23](#). *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

Table D.4. The Great Chicago and Boston Fires and Non-Wood Construction Manufacturing, Entire United States

	Number of Establishments	Production Value	Labor Costs	Material Costs	Capital
	(1)	(2)	(3)	(4)	(5)
Panel A: Cook County					
Cook County $\times$ Post-1871	1.113*** (0.157)	1.733** (0.712)	1.652*** (0.636)	1.248* (0.645)	0.945 (0.704)
Mean Dep. Var.	1.229	6.486	5.758	5.699	5.988
Panel B: Suffolk County					
Suffolk County $\times$ Post-1871	0.283* (0.159)	1.412* (0.799)	1.085 (0.711)	1.243* (0.693)	1.271 (0.878)
Mean Dep. Var.	1.229	6.486	5.758	5.700	5.988
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
# Cities	751	751	751	751	751
# Observations	2,236	2,236	2,236	2,236	2,236

Notes. This Table reports the effect of the Great Chicago (1871) and Boston (1872) Fires on non-wood construction activity. The unit of observation is a county observed at a decade frequency between 1860 and 1880. The sample includes all counties in the United States. The dependent variable is: the number of establishments (1), production value (2), labor costs (3), material costs (4), and capital (5). All dependent variables refer to the non-wood industries and are expressed in log terms. The treatment is an interaction term equal to one for Cook County (where Chicago is located, in Panel A) or Suffolk County (where Boston is located, in Panel B) in 1880, after their respective fires, and zero otherwise. All regressions are estimated through OLS. Regressions include county and year fixed effects. Standard errors are clustered at the county level. Referenced on pages: 20, 23. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

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